

BINOMIAL THEOREM

JEE EXERCISES

Exercise J-I

BASED ON IMPORTANT TERMS

1. If $\frac{T_2}{T_3}$ in the expansion of $(a+b)^n$ and $\frac{T_3}{T_4}$ in the expansion of $(a+b)^{n+3}$ are equal, then n is equal to :

यदि $(a+b)^n$ के प्रसार में $\frac{T_2}{T_3}$ तथा $(a+b)^{n+3}$ के प्रसार में $\frac{T_3}{T_4}$ बराबर है तब n का मान होगा:

- (A) 3 (B) 4 (C) 5 (D) 6

Ans. C

Sol. $\frac{T_2}{T_3}(a+b)^n = \sum_{r=0}^n {}^nC_r a^{n-r} b^r$

$$\Rightarrow \frac{T_2}{T_3} = \frac{{}^nC_1 a^{n-1} b^1}{{}^nC_2 a^{n-2} b^2} = \frac{|n|2|n-2|}{|n-1|n} \cdot \frac{a}{b} = \frac{2a}{(n-1)b}$$

$$(a+b)^{n+3} = \sum_{r=0}^{n+3} {}^{n+3}C_r a^{n+3-r} b^r$$

$$\Rightarrow \frac{T_3}{T_4} = \frac{{}^{n+3}C_2 a^{n+3-2} b^2}{{}^{n+3}C_3 a^{n+3-3} b^3} = \frac{|n+3|n|3|}{|n+1|2|n+3|} \cdot \frac{a^{n+1} \cdot b^2}{a^n b^3}$$

$$= \frac{3}{n+1} \cdot \frac{a}{b}$$

Given $\frac{T_2}{T_3} = \frac{T_3}{T_4}$

$$\Rightarrow \frac{2a'}{(n-1)b'} = \frac{3a'}{(n+1)b'}$$

$$\Rightarrow 2n+2 = 3n-3$$

$$\Rightarrow n = 5$$

2. If number of terms in the expansion of $(x-2y+3z)^n$ are 45, then n is equal to :

- (A) 7 (B) 8 (C) 9 (D) None of these

यदि $(x - 2y + 3z)^n$ के प्रसार में कुल पदों की संख्या 45 है तब n का मान होगा :

- (A) 7 (B) 8 (C) 9 (D) इनमें से कोई नहीं

Ans. B

Sol. No. of terms in the expansion of $(x - 2y + 3z)^n$ are 45

$$(x - 2y + 3z)^n = \sum_{r_1+r_2+r_3=n} \frac{n!}{r_1!r_2!r_3!} \cdot x^{r_1} \cdot (-2)^{r_2} \cdot y^{r_2} \cdot 3^{r_3} \cdot z^{r_3}$$

$$\Rightarrow n + {}^{3-1}C_{3-1} = 45 \Rightarrow {}^{n+2}C_2 = 45 \Rightarrow n^2 + 3n + 2 = 90$$

$$\Rightarrow n = -11 \text{ or } n = 8$$

3. The greatest coefficient in the expansion of $(1 + x)^{2n+2}$ is :

$(1 + x)^{2n+2}$ के प्रसार में महत्तम पद का गुणांक होगा :

- (A) $\frac{(2n)!}{(n!)^2}$ (B) $\frac{(2n+2)!}{[(n+1)!]^2}$ (C) $\frac{(2n+2)!}{n!(n+1)!}$ (D) $\frac{(2n)!}{n!(n+1)!}$

Ans. B

Sol. $(1 + x)^{2n+2} = \sum_{r=0}^{2n+2} {}^{2n+2}C_r x^r$

${}^{2n+2}C_r$ is maximum if $r = n + 1$

$$\Rightarrow \text{Greatest coefficient is } {}^{2n+2}C_{n+1} = \frac{(2n+2)!}{(n+1)!(n+1)!}$$

4. The expression $\frac{1}{\sqrt{4x+1}} \left[\left[\frac{1+\sqrt{4x+1}}{2} \right]^7 - \left[\frac{1-\sqrt{4x+1}}{2} \right]^7 \right]$ is a polynomial in x of degree :

व्यंजक $\frac{1}{\sqrt{4x+1}} \left[\left[\frac{1+\sqrt{4x+1}}{2} \right]^7 - \left[\frac{1-\sqrt{4x+1}}{2} \right]^7 \right]$, x में किस घात का बहुपदीय व्यंजक है :

- (A) 7 (B) 5 (C) 4 (D) 3

Ans. D

Sol. Given expression can be rewritten as

$$\frac{2}{2^7 \sqrt{4x+1}} \left[{}^7C_1(\sqrt{4x+1}) + {}^7C_3(\sqrt{4x+1})^3 + \dots + {}^7C_7(\sqrt{4x+1})^7 \right]$$

$$\frac{1}{2^6} \left[{}^7C_1 + {}^7C_3(\sqrt{4x+1})^2 + \dots + {}^7C_7(\sqrt{4x+1})^6 \right]$$

$\therefore$ Last term becomes $(4x+1)^3$

Hence degree is 3

5. If the coefficient of $(r+2)^{\text{th}}$ and $(3r)^{\text{th}}$ term in the exp. of $(1+x)^{2n}$ are equal then :

(A) $n = 2r + 1$ (B) $n = 2r - 1$ (C) $n = 2r$ (D) None of these

$(1+x)^{2n}$ के प्रसार में $(r+2)^{\text{वें}}$ तथा $(3r)^{\text{वें}}$ पद के गुणांक बराबर हो तो :

(A) $n = 2r + 1$ (B) $n = 2r - 1$ (C) $n = 2r$; व द्वाइनमें से कोई नहीं

Ans. C

Sol. $(r+2)^{\text{th}} = (3r)^{\text{th}}$

$${}^{2n}C_{r+2} = {}^{2n}C_{3r}$$

$$r+2 = 3r \quad \text{or} \quad r+2+2r = 2n$$

$$r = 1 \quad \text{or} \quad n = 2r + 1$$

6. The coefficient of the middle term in the binomial expansion in powers of x of $(1+\alpha x)^4$ and of $(1-\alpha x)^6$ is the same if α equals :

$(1+\alpha x)^4$ तथा $(1-\alpha x)^6$ के द्विपद प्रसार, जो कि x की घातों के रूप में है, के मध्य पदों के गुणांक समान होंगे यदि α का मान है :

(A) $-\frac{5}{3}$ (B) $\frac{10}{3}$ (C) $-\frac{3}{10}$ (D) $\frac{3}{5}$

Ans. C

Sol. Coefficient of middle term in $(1+\alpha x)^4 = {}^4C_2 \alpha^2$

coefficient of middle term in $(1-\alpha x)^6 = {}^6C_3 (-\alpha)^3$

$${}^4C_2 \alpha^2 = - {}^6C_3 \alpha^3$$

$$-\frac{6}{20} = \alpha$$

$$a = \frac{-3}{10}$$

7. If the coefficients of r^{th} , $(r+1)^{\text{th}}$ and $(r+2)^{\text{th}}$ terms in the binomial expansion of $(1+y)^m$ are in A.P., then m and r satisfy the equation :

एक द्विपद $(1+y)^m$ के प्रसार में यदि r वें, $(r+1)$ वें तथा $(r+2)$ वें पदों के गुणांक समान्तर श्रेणी में हैं, तो m तथा r निम्न समीकरण को संतुष्ट करेंगे

$$(A) m^2 - m(4r - 1) + 4r^2 - 2 = 0$$

$$(B) m^2 - m(4r + 1) + 4r^2 + 2 = 0$$

$$(C) m^2 - m(4r + 1) + 4r^2 - 2 = 0$$

$$(D) m^2 - m(4r - 1) + 4r^2 + 2 = 0$$

Ans. C

Sol. Coefficient of r^{th} , $(r + 1)^{\text{th}}$ and $(r + 2)^{\text{th}}$ terms in expansion of $(1 + y)^n$ are ${}^m C_{r-1}$, ${}^m C_r$, ${}^m C_{r+1}$

$$\text{Then } 2 \cdot {}^m C_r = {}^m C_{r-1} + {}^m C_{r+1}$$

$$\Rightarrow m^2 - m(4r + 1) + 4r^2 - 2 = 0$$

Short Method: Let $r = 1$, hence ${}^m C_0$, ${}^m C_1$ and ${}^m C_2$ are in A.P.

$$\Rightarrow 2 \cdot {}^m C_1 = {}^m C_0 + {}^m C_2 \Rightarrow 2m = 1 + \frac{m(m-1)}{2}$$

$$\Rightarrow 4m = 2 + m^2 - m \Rightarrow m^2 - 5m + 2 = 0$$

Which is given by (C)

8. If the coefficient of x^7 in $\left[ax^2 + \left(\frac{1}{bx}\right)\right]^{11}$ equals the coefficient of x^{-7} in $\left[ax - \left(\frac{1}{bx^2}\right)\right]^{11}$, then a and b satisfy the relation :

यदि $\left[ax^2 + \left(\frac{1}{bx}\right)\right]^{11}$ के प्रसार में x^7 का गुणांक $\left[ax - \left(\frac{1}{bx^2}\right)\right]^{11}$ के प्रसार में x^{-7} के गुणांक के बराबर है, तो a और b में सम्बन्ध है :

$$(A) a - b = 1$$

$$(B) a + b = 1$$

$$(C) \frac{a}{b} = 1$$

$$(D) ab = 1$$

Ans. D

Sol. In the expansion of $\left(ax^2 + \frac{1}{bx}\right)^{11}$, the general term is

$$T_{r+1} = {}^{11}C_r (ax^2)^{11-r} \left(\frac{1}{bx}\right)^r = {}^{11}C_r a^{11-r} \frac{1}{b^r} x^{22-3r}$$

For x^7 , we must have $22 - 3r = 7 \Rightarrow r = 5$, and the coefficient of $x^7 = {}^{11}C_5 \cdot a^{11-5} \frac{1}{b^5} = {}^{11}C_5 \frac{a^6}{b^5}$.

Similarly, in the expansion of $\left(ax^2 - \frac{1}{bx}\right)^{11}$, the general term is $T_{r+1} = {}^{11}C_r (-1)^r \frac{a^{11-r}}{b^r} x^{11-3r}$

For x^{-7} we must have $11 - 3r = -7 \Rightarrow r = 6$, and the coefficient of x^{-7} is ${}^{11}C_6 \frac{a^5}{b^6} = {}^{11}C_5 \frac{a^5}{b^6}$

9. Let x be the 7th term from the beginning and y be the 7th term from the end in the expansion of $\left(3^{1/3} + \frac{1}{4^{1/3}}\right)^n$. If y = 12x then the value of n is :

माना $\left(3^{1/3} + \frac{1}{4^{1/3}}\right)^n$ के प्रसार में शुरू से 7वाँ पद x तथा अंत से 7वाँ पद y है। यदि y = 12x है, तो n बराबर होगा:

- (A) 9 (B) 8 (C) 10 (D) 11

Ans. A

Sol.
$$\left(3^{1/3} + \frac{1}{4^{1/3}}\right)^n = \sum_{r=0}^n {}^nC_r 3^{\frac{n-r}{3}} \cdot 4^{-\frac{r}{3}}$$

$$x = {}^nC_6 3^{\frac{n-6}{3}} \cdot 4^{-\frac{6}{3}} \quad \& \quad y = {}^nC_{n-6} 3^{\frac{n-(n-6)}{3}} \cdot 4^{-\frac{(n-6)}{3}}$$

Given y = 12x

$$\Rightarrow {}^nC_6 3^2 \cdot 4^{-\frac{(n-6)}{3}} = {}^nC_6 3^{\frac{n-6}{3}} \cdot 4^{-2} \cdot 4 \cdot 3$$

$$\Rightarrow 4^{-\frac{(n-6)}{3}+2} = 3^{\frac{n-6}{3}-2} \cdot 4 \cdot 3$$

$$\Rightarrow 4^{\frac{-n+6+6}{3}} = 3^{\frac{n-12}{3}} \cdot 4 \cdot 3$$

$$\Rightarrow 4^{\frac{-n+12-3}{3}} = 3^{\frac{n-12+3}{3}}$$

This is possible if n = 9

10. In the binomial expansion of $(a - b)^n$, $n \geq 5$, the sum of 5th and 6th terms is zero, then $\frac{a}{b}$ equals :

$(a - b)^n$, $n \geq 5$ के द्विपद प्रसार में, 5^{वें} तथा 6^{वें} पदों का योग 0 है, तब $\frac{a}{b}$ बराबर है:

- (A) $\frac{5}{n-4}$ (B) $\frac{6}{n-5}$ (C) $\frac{n-5}{6}$ (D) $\frac{n-4}{5}$

Ans. D

Sol. $(a-b)^n \Rightarrow$ general term $T_{r+1} = {}^nC_r \cdot a^{n-r} \cdot b^r (-1)^r$
 ${}^nC_4 \cdot a^{n-4} \cdot b^4 - {}^nC_5 \cdot a^{n-5} \cdot b^5 = 0$

$${}^nC_4 \left(\frac{a}{b} \right) = {}^nC_5 \Rightarrow \frac{n!}{4!(n-4)!} \left(\frac{a}{b} \right) = \frac{n!}{5!(n-5)!}$$

$$\left(\frac{a}{b} \right) = \frac{(n-4)}{5}$$

BASED ON FINDING COEFFICIENTS

11. If $(1 + ax)^n = 1 + 8x + 24x^2 + \dots$ then the value of a and n is :

यदि $(1 + ax)^n = 1 + 8x + 24x^2 + \dots$ तब a तथा n का मान होगा:

- (A) 2, 4 (B) 2, 3 (C) 3, 6 (D) 1, 2

Ans. A

Sol. $(1 + ax)^n = 1 + 8x + 24x^2 + \dots$

$$\Rightarrow 1 + {}^nC_1(ax) + {}^nC_2(ax)^2 + \dots = 1 + 8x + 24x^2 + \dots$$

$$\Rightarrow na = 8 \text{ \& } \frac{n}{n-2} a^2 = 24$$

$$\Rightarrow na = 8 \text{ \& } n(n-1)a^2 = 48$$

$$\Rightarrow na = 8 \text{ \& } na(na-a) = 48$$

$$8(8-a) = 48$$

$$\Rightarrow 8-a = 6$$

$$\Rightarrow a = 2$$

$$\Rightarrow n = 4$$

12. The coefficient of x^4 in the expansion of $(1 + x + x^2 + x^3)^n$ is :

$(1 + x + x^2 + x^3)^n$ के प्रसार में x^4 को गुणांक होगा :

- (A) nC_4 (B) ${}^nC_4 + {}^nC_2$ (C) ${}^nC_4 + {}^nC_2 + {}^nC_4 \cdot {}^nC_2$ (D) ${}^nC_4 + {}^nC_2 + {}^nC_1 \cdot {}^nC_2$

Ans. D

Sol. $(1 + x + x^2 + x^3)^n = ((1 + x^2)(1 + x))^n$

$$= (1+x^2)^n (1+x)^n = \left[\sum_{r=0}^n {}^nC_r x^{2r} \right] \left[\sum_{r=0}^n {}^nC_r x^r \right]$$

Coefficient of x^4

$$= {}^nC_0 {}^nC_4 + {}^nC_1 {}^nC_2 + {}^nC_2 \cdot {}^nC_0$$

$$= {}^nC_4 + {}^nC_2 + {}^nC_1 \cdot {}^nC_2$$

13. The co-efficient of x^9 in the polynomial given by $(x+1)(x+2) \dots (x+10) + (x+2)(x+3) \dots (x+11) + \dots + (x+11)(x+12) \dots (x+20)$ is :

व्यंजक $(x+1)(x+2) \dots (x+10) + (x+2)(x+3) \dots (x+11) + \dots + (x+11)(x+12)$

में x^9 का गुणांक है:

- (A) 5511 (B) 5151 (C) 1515 (D) 1155

Ans. D

Sol. Coefficient of x^9 in $(x+1)(x+2) \dots (x+10) + (x+2) \dots (x+11) + \dots + (x+11)(x+12) \dots (x+20)$

Let $P(x) = (x+1)(x+2) \dots (x+10)$

$\Rightarrow x = -1, \dots, -10$ are roots of $P(x)$

Coefficient of $x^9 = -(-1 - 2 \dots - 10) = 1 + \dots + 10$

$$= \frac{10 \times 11}{2} = 55$$

Coefficient of x^9

$= 55 + 65 + 75 + 85 + 95 + 105 + 115 + 125 + 135 + 145 + 155 \rightarrow$ A.P. with $d = 10$

$$= \frac{11}{2} [110 + 10 \times 10] = \frac{210}{2} \times 11$$

$$= 105 \times 11 = 1155$$

14. The coefficient of x^{39} in the expansion of $\left(x^4 - \frac{1}{x^3}\right)^{15}$ is :

$\left(x^4 - \frac{1}{x^3}\right)^{15}$ के प्रसार में x^{39} का गुणांक है:

(A) - 455

(B) - 105

(C) + 455

(D) + 105

Ans. A

Sol.
$$\left(x^4 - \frac{1}{x^3}\right)^{15} = \sum_{r=0}^{15} {}^{15}C_r (x^4)^{15-r} \left(\frac{-1}{x^3}\right)^r$$

$$= \sum_{r=0}^{15} {}^{15}C_r (-1)^r x^{60-7r}$$

For x^{39} ; $60 - 7r = 39$

$$\Rightarrow 7r = 60 - 39$$

$$\Rightarrow r = 3$$

Coefficient of x^{39} is $(-1)^3 \cdot {}^{15}C_3$

$$= -455$$

15. The coefficient of x^n in expansion of $(1+x)(1-x)^n$ is : $(1+x)(1-x)^n$ के प्रसार में x^n का गुणांक होगा :

(A) $(n-1)$

(B) $(-1)^n (1-n)$

(C) $(-1)^{n-1} (n-1)^2$

(D) $(-1)^{n-1} n$

Ans. B**Sol.** Coefficient of x^n in expansion of $(1+x)(1-x)^n$ i.e., coefficient of x^n in expansion of $(1-x)^n$ + coefficient of x^{n-1} in expansion of $(1-x)^n$]

Now, $(-1)^n {}^nC_n + (-1)^{n-1} {}^nC_{n-1}$

$$(-1)^n [{}^nC_n - {}^nC_{n-1}] = (-1)^n [1-n]$$

16. For natural numbers m, n if $(1-y)^m (1+y)^n = 1 + a_1 y + a_2 y^2 + \dots$, and $a_1 = a_2 = 10$, then (m, n) is :प्राकृत संख्याओं m, n के लिए यदि $(1-y)^m (1+y)^n = 1 + a_1 y + a_2 y^2 + \dots$, तथा $a_1 = a_2 = 10$, तो (m, n) है :

(A) (35, 20)

(B) (45, 35)

(C) (35, 45)

(D) (20, 45)

Ans. C

Sol. $(1-y)^m (1+y)^n = 1 + a_1 y + a_2 y^2 + \dots$

$$(1-my + {}^mC_2 y^2 \dots) (1+ny + {}^nC_2 y^2 \dots) = 1 + a_1 y + a_2 y^2 + \dots$$

$$a_1 = n - m = 10 \quad \dots (1)$$

$$a_2 = {}^mC_2 + {}^nC_2 - mn = 10 \quad \dots (2)$$

solving (1) & (2), we get $(m, n) \equiv (35, 45)$

17. The coefficient of x^7 in the expansion of $(1 - x - x^2 + x^3)^6$ is :

[AIEEE 2011]

$(1 - x - x^2 + x^3)^6$ के प्रसार में x^7 का गुणांक है :

- (A) -144 (B) 132 (C) 144 (D) -132

Ans. A

Sol. $(1 - x - x^2 + x^3)^6 \Rightarrow [(1 - x) - x^2(1 - x)]$

$$\Rightarrow (1 - x)^6 (1 - x^2)^6$$

$$\Rightarrow (1 - {}^6C_1 x + {}^6C_2 x^2 - {}^6C_3 x^3 + {}^6C_4 x^4 - {}^6C_5 x^5 + {}^6C_6 x^6) \times (1 - {}^6C_1 x^2 + {}^6C_2 x^4 - {}^6C_3 x^6 + \dots)$$

$\Rightarrow$ coefficient of x^7 is $\rightarrow$

$${}^6C_1 \cdot {}^6C_3 - {}^6C_3 \cdot {}^6C_2 + {}^6C_5 \cdot {}^6C_1$$

$$\Rightarrow \frac{6.5.4}{3.2} - \frac{6.5.4}{3.2.1} + 6.6 = 120 - 300 + 36$$

$$\Rightarrow -144$$

18. If the coefficient of x^3 and x^4 in the expansion of $(1 + ax + bx^2)(1 - 2x)^{18}$ in powers of x are both zero then (a, b) is equal to :

[JEE(Main) 2014]

यदि $(1 + ax + bx^2)(1 - 2x)^{18}$ के x की घातों में प्रसार में x^3 तथा x^4 दोनों के गुणांक शून्य हैं, तो (a, b) बराबर है :

- (A) $\left(14, \frac{272}{3}\right)$ (B) $\left(16, \frac{272}{3}\right)$ (C) $\left(16, \frac{251}{3}\right)$ (D) $\left(14, \frac{251}{3}\right)$

Ans. B

Sol. $(1 + ax + bx^2)(1 - 2x)^{18}$

$$\text{coeff of } x^3 = {}^{18}C_3(-2)^3 + a(-2)^2 \cdot {}^{18}C_2 + b(-2) \cdot {}^{18}C_1 = 0$$

$$\text{coeff of } x^4 = {}^{18}C_4(-2)^4 + a(-2)^3 \cdot {}^{18}C_3 + b(-2)^2 \cdot {}^{18}C_2 = 0$$

$$\Rightarrow 51a - 3b = 544 \text{ and } 32a - 3b = 240$$

Subtracting we get $a = 16$

$$\Rightarrow b = \frac{272}{3}$$

REMAINDER BASED PROBLEMS

19. The number 7^{1995} when divided by 100 leaves the remainder :

- (A) 43 (B) 45 (C) 50 (D) None of these

जब संख्या 7^{1995} को 100 से विभाजित किया जाता है तब शेषफल होगा

- (A) 43 (B) 45 (C) 50 ; क द्धनमें से कोई नहीं

Ans. A

Sol. $7^{1995} = 7[(7)^2]^{997} = 7(50 - 1)^{997}$

using expansion

$$7\{^{997}C_0 \cdot 50^{997} + ^{997}C_1 \cdot 50^{996} + \dots + ^{997}C_{996} \cdot 50 - 1\}$$

$$= 100K + 7(997 \times 50 - 1)$$

$$= 100K + 7(990 \times 50 + 7 \times 50 - 1)$$

$$= 100K_1 + 49 \times 50 - 7 = 100K_1 + 2450 - 7$$

$$= 100K_1 + 2400 + 43$$

$\therefore$ Remainder is 43

20. The last digit of $(3^P + 2)$, where $P = 3^{4n+2}$ is :

$(3^P + 2)$ का अन्तिम अंक क्या होगा जबकि $P = 3^{4n+2}$:

(A) 1

(B) 9

(C) 4

(D) 5

Ans. D

Sol. $(3^P + 2)$; where $P = 3^{4n+2}$

$$P = 3^{4n} \cdot 9 = 81^n \cdot 9$$

$$P = 9 \cdot (80 + 1)^n$$

$$= 9 \sum_{r=0}^n {}^nC_r (80)^{n-r}$$

$$P = 10\lambda + 9 \cdot {}^nC_n$$

$$\Rightarrow \text{Remainder} = 9 \text{ last digit of } (3^{3^{4n+2}} + 2) = \text{last digit of } 3^9 + 2$$

$$= 9^4 \cdot 3 + 2$$

$$= (10 - 1)^4 \cdot 3 + 2$$

$$= 3 + 2 + 10\lambda'$$

$$\backslash \text{Remainder} = 5$$

21. The remainder left out when $8^{2n} - (62)^{2n+1}$ is divided by 9 is :

[AIEEE-2009]

$8^{2n} - (62)^{2n+1}$ को 9 से भाग देने पर शेषफल बचता है :

(A) 0

(B) 2

(C) 7

(D) 8

Ans. B

Sol. $8^{2n} - (62)^{2n+1}$
 $= (9 - 1)^{2n} - (63 - 1)^{2n+1}$
 $= \sum_{r=0}^{2n} {}^{2n}C_r 9^{2n-r} (-1)^r - \sum_{r=0}^{2n+1} {}^{2n+1}C_r (63)^{2n+1-r} (-1)^r$
 $= 9\lambda + 1 + 1$
 $= 9\lambda + 2$
 $\therefore \text{Remainder} = 2$

22. If $X = \{4^n - 3n - 1 : n \in \mathbb{N}\}$ and $Y = \{9(n-1) : n \in \mathbb{N}\}$, where $\mathbb{N}$ is the set of natural numbers, then

$X \cup Y$ is equal to :

[JEE(Main) 2014]

यदि $X = \{4^n - 3n - 1 : n \in \mathbb{N}\}$ तथा $Y = \{9(n-1) : n \in \mathbb{N}\}$, हैं, जहाँ $\mathbb{N}$ प्राकृत संख्याओं का समुच्चय है, तो

$X \cup Y$ बराबर है :

- (A) X (B) Y (C) $\mathbb{N}$ (D) $Y - X$

Ans. B

Sol. $4^n - 3n - 1$
 $= (1 + 3)^n - 3n - 1$
 $= {}^nC_2 3^2 + {}^nC_3 3^3 + \dots$
 $= 9[{}^nC_2 + 3 \times {}^nC_3 + \dots]$

Set X contains numbers which are multiples of 9 but not all multiples of 9

$\therefore X \cup Y = Y$

NUMERICALLY GREATEST TERM

23. Only 4th term is numerically greatest term in $\left(2 + \frac{3x}{8}\right)^{10}$. Then the range of x is :

$\left(2 + \frac{3x}{8}\right)^{10}$ के प्रसार में केवल 4th पद महत्तम पद हो तो x का परिसर है :

- (A) $\left(\frac{-64}{21}, -2\right) \cup \left(1, \frac{64}{25}\right)$ (B) $\left(\frac{-64}{21}, -2\right) \cup \left(2, \frac{64}{21}\right)$
 (C) $\left(-2, \frac{-64}{21}\right) \cup \left(2, \frac{64}{21}\right)$ (D) $\left(\frac{-64}{21}, -2\right) \cup \left(1, \frac{64}{21}\right)$

Ans. B

Sol. $\left(2 + \frac{3x}{8}\right)^{10}$

T_4 is only NGT

$$\Rightarrow [P] = 3 \text{ \& } P \notin \mathbb{Z}$$

$$\Rightarrow 3 < P < 4$$

$$\Rightarrow 3 < \frac{11}{1 + \left\lfloor \frac{16}{3x} \right\rfloor} < 4$$

$$\Rightarrow 3 \left(1 + \left\lfloor \frac{16}{3x} \right\rfloor \right) - 11 < 0$$

&

$$11 < 4 \left(1 + \left\lfloor \frac{16}{3x} \right\rfloor \right)$$

$$\Rightarrow \frac{16}{|x|} < 8$$

$$7 < \frac{64}{3|x|}$$

$$\Rightarrow |x| > 2 \{ \because |x| > 0 \}$$

$$\Rightarrow |x| < \frac{64}{21}$$

$$\Rightarrow x \in (-\infty, -2) \cup (2, \infty)$$

$$x \in \left(-\frac{64}{21}, 0 \right) \cup \left(0, \frac{64}{21} \right)$$

$$\Rightarrow x \in \left(-\frac{64}{21}, -2 \right) \cup \left(2, \frac{64}{21} \right)$$

INTEGRAL PART AND FRACTIONAL PART BASED CONCEPT

24. If n is a positive integer, then $(\sqrt{3} + 1)^{2n} - (\sqrt{3} - 1)^{2n}$ is :

[AIEEE-2012]

(A) A rational number other than positive integers

(B) An irrational number

(C) An odd positive integer

(D) An even positive integer

यदि n एक धनपूर्णांक है, तो $(\sqrt{3}+1)^{2n} - (\sqrt{3}-1)^{2n}$:

[AIEEE-2012]

- (A) धनपूर्णाकों को छोड़कर एक परिमेय संख्या है (B) एक अपरिमेय संख्या है
(C) एक विषम धनपूर्णांक है (D) एक सम धनपूर्णांक है।

Ans. B

Sol. $(\sqrt{3}+1)^{2n} - (\sqrt{3}-1)^{2n}$
 $= 2[T_2 + T_4 + T_6 + \dots + T_{2n}]$
 $= 2[{}^{2n}C_1(\sqrt{3})^{2n-1} + {}^{2n}C_3(\sqrt{3})^{2n-3} + \dots]$
 $= \text{An Irrational Number}$

SUMMATION OF SERIES

25. $\sum_{r=1}^n r \cdot {}^{2n}C_r$ is equal to :

$\sum_{r=1}^n r \cdot {}^{2n}C_r$ का मान है :

- (A) $n \cdot 2^{n-1}$ (B) 2^{2n-1} (C) $2^{n-1} + 1$ (D) $n \cdot 2^{2n-1}$

Ans. D

Sol. $\sum_{r=1}^n r \cdot {}^{2n}C_r = \sum_{r=1}^n r \cdot \frac{2n}{r} \cdot {}^{2n-1}C_{r-1}$
 $= 2n \sum_{r=1}^n {}^{2n-1}C_{r-1}$
 $= 2n \left({}^{2n-1}C_0 + {}^{2n-1}C_1 + \dots + {}^{2n-1}C_{n-1} \right)$
 $= 2 \cdot n$
 $= 2n \cdot \frac{2^{2n-1}}{2} = n \cdot 2^{2n-1}$

26. If $C_0, C_1, C_2, \dots, C_n$ denote the coefficients in the expansion of $(1+x)^n$, then the value of $\sum_{r=0}^n (r+1)C_r$ is :

यदि $(1+x)^n$ के प्रसार में $C_0, C_1, C_2, \dots, C_n$ द्विपद गुणांक है तब $\sum_{r=0}^n (r+1)C_r$ है :

(A) $n \cdot 2^n$

(B) $(n+1)2^{n-1}$

(C) $(n+2)2^{n-1}$

(D) $(n+2)2^{n-2}$

Ans. C

Sol.
$$\sum_{r=0}^n (r+1)C_r = \sum_{r=0}^n rC_r + \sum_{r=0}^n C_r$$

$$= n \sum_{r=1}^n {}^{n-1}C_{r-1} + 2^n$$

$$= n \cdot 2^{n-1} + 2^n$$

$$= 2^{n-1} (n+2)$$

27. If $(1+x)^n = C_0 + C_1 x + C_2 x^2 + \dots + C_n x^n$, then $C_0 + 5C_1 + 9C_2 + \dots + (4n+1)C_n$ is equal to :

यदि $(1+x)^n = C_0 + C_1 x + C_2 x^2 + \dots + C_n x^n$, तब $C_0 + 5C_1 + 9C_2 + \dots + (4n+1)C_n$ बराबर होगा :

(A) $n \cdot 2^n$

(B) $(n+1)2^n$

(C) $(2n+1)2^n$

(D) $(4n+1)2^n$

Ans. C

Sol. $C_0 + {}^5C_1 + {}^9C_2 + \dots + (4n+1)C_n$

$$= \sum_{r=0}^n (4r+1)C_r$$

$$= 4 \sum_{r=0}^n rC_r + \sum_{r=0}^n C_r$$

$$= 4 \sum_{r=1}^n r \cdot \frac{n}{r} {}^{n-1}C_{r-1} + \sum_{r=0}^n C_r$$

$$= 4n \cdot 2^{n-1} + 2^n$$

$$= (4n+2)2^{n-1}$$

$$= (2n+1)2^n$$

28. If $(1-x+x^2)^n = a_0 + a_1 x + a_2 x^2 + \dots + a_{2n} x^{2n}$, then $a_0 + a_2 + a_4 + \dots + a_{2n}$ equals :

यदि $(1-x+x^2)^n = a_0 + a_1 x + a_2 x^2 + \dots + a_{2n} x^{2n}$, तब $a_0 + a_2 + a_4 + \dots + a_{2n}$ बराबर होगा:

(A) $\frac{3^n+1}{2}$

(B) $\frac{3^n-1}{2}$

(C) $\frac{1-3^n}{2}$

(D) $3^n + \frac{1}{2}$

Ans. A

Sol. $(1 - x + x^2)^n = a_0 + a_1x + \dots + a_{2n}x^{2n}$

Put $x = 1$

$$1 = a_0 + a_1 + \dots + a_{2n} \quad \dots(i)$$

Put $x = -1$

$$3^n = a_0 - a_1 + \dots + a_{2n} \quad \dots(ii)$$

(i) + (ii)

$$2[a_0 + a_2 + \dots + a_{2n}] = 3^n + 1$$

$$\Rightarrow a_0 + a_2 + \dots + a_{2n} = \frac{3^n + 1}{2}$$

29. If $(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$, then $\frac{C_1}{C_0} + \frac{2C_2}{C_1} + \frac{3C_3}{C_2} + \dots + \frac{nC_n}{C_{n-1}} =$

यदि $(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$, तो $\frac{C_1}{C_0} + \frac{2C_2}{C_1} + \frac{3C_3}{C_2} + \dots + \frac{nC_n}{C_{n-1}} =$

(A) $\frac{n}{2}$ (B) $n(n+1)$ (C) $\frac{n(n+1)}{12}$ (D) $\frac{n(n+1)}{2}$

Ans. D

Sol. $\frac{C_1}{C_0} + \frac{2C_2}{C_1} + \frac{3C_3}{C_2} + \dots + \frac{n \cdot C_n}{C_{n-1}}$

$$= \sum_{r=1}^n r \cdot \frac{C_r}{C_{r-1}}$$

$$\left\{ \because \frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r} \right\}$$

$$= \sum_{r=1}^n r \left(\frac{n-r+1}{r} \right) = \sum_{r=1}^n (n-r+1)$$

$$= (n+1) \sum_{r=1}^n 1 - \sum_{r=1}^n r$$

$$= n(n+1) - \frac{n(n+1)}{2}$$

$$= \frac{n(n+1)}{2}$$

30. If $S_n = \sum_{r=0}^n \frac{1}{{}^nC_r}$ and $t_n = \sum_{r=0}^n \frac{r}{{}^nC_r}$, then $\frac{t_n}{s_n}$ is equal to :

यदि $S_n = \sum_{r=0}^n \frac{1}{{}^nC_r}$ एवं $t_n = \sum_{r=0}^n \frac{r}{{}^nC_r}$, तो $\frac{t_n}{s_n}$ बराबर है :

- (A) $\frac{1}{2}n$ (B) $\frac{1}{2}n-1$ (C) $n-1$ (D) $\frac{2n-1}{2}$

Ans. A

Sol. $s_n = \sum_{r=0}^n \frac{1}{{}^nC_r}$

$$t_n = \sum_{r=0}^n \frac{r}{{}^nC_r}$$

$$t_n = \sum_{r=0}^n \frac{n-r}{{}^nC_r} \quad \therefore \quad 2t_n = n \sum_{r=0}^n \frac{1}{{}^nC_r} = ns_n \quad \Rightarrow \quad \frac{t_n}{s_n} = \frac{n}{2}$$

31. The sum of the series ${}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \dots - \dots + {}^{20}C_{10}$ is:
श्रेणी, ${}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \dots - \dots + {}^{20}C_{10}$ का योग है :

- (A) $- {}^{20}C_{10}$ (B) $\frac{1}{2} {}^{20}C_{10}$ (C) 0 (D) ${}^{20}C_{10}$

Ans. B

Sol. $s = {}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \dots + {}^{20}C_{10}$

$$s = {}^{20}C_{10} - {}^{20}C_{11} + \dots + {}^{20}C_{20}$$

$$2s = 0 + {}^{20}C_{10}$$

$$s = \frac{1}{2} ({}^{20}C_{10})$$

32. The sum of coefficients of integral powers of x in the binomial expansion of $(1-2\sqrt{x})^{50}$ is :

$(1-2\sqrt{x})^{50}$ के द्विपद प्रसार में x को पूर्णांकीय घातों के गुणांकों का योग है :

[JEE(Main) 2015]

- (A) $\frac{1}{2}(3^{50}+1)$ (B) $\frac{1}{2}(3^{50})$ (C) $\frac{1}{2}(3^{50}-1)$ (D) $\frac{1}{2}(2^{50}+1)$

Ans. A

Sol. $(1 - 2\sqrt{x})^{50} = 1 - {}^{50}C_1(2\sqrt{x}) + {}^{50}C_2(x^2)(\sqrt{x})^2 + \dots$

Put $\sqrt{x} = 1$ & $\sqrt{x} = -1$ & then add

Ans, $\frac{3^{50} + 1}{2}$

33. The value of $({}^{21}C_1 - {}^{10}C_1) + ({}^{21}C_2 - {}^{10}C_2) + ({}^{21}C_3 - {}^{10}C_3) + ({}^{21}C_4 - {}^{10}C_4) + \dots + ({}^{21}C_{10} - {}^{10}C_{10})$ is :

${}^{21}C_1 - {}^{10}C_1 + ({}^{21}C_2 - {}^{10}C_2) + ({}^{21}C_3 - {}^{10}C_3) + ({}^{21}C_4 - {}^{10}C_4) + \dots + ({}^{21}C_{10} - {}^{10}C_{10})$ का मान है -

[JEE(Main) 2017]

- (A) $2^{20} - 2^9$ (B) $2^{20} - 2^{10}$ (C) $2^{21} - 2^{11}$ (D) $2^{21} - 2^{10}$

Ans. B

Sol. $({}^{21}C_0 + {}^{21}C_1 + {}^{21}C_2 + \dots + {}^{21}C_{10}) - ({}^{10}C_0 + {}^{10}C_1 + \dots + {}^{10}C_{10}) - ({}^{21}C_0 - {}^{10}C_0)$

$$\frac{2^{21}}{2} - 2^{10} = 2^{20} - 2^{10}$$

34. The sum of the co-efficients of all odd degree terms in the expansion of $(x + \sqrt{x^3 - 1})^5 + (x - \sqrt{x^3 - 1})^5$,

$(x > 1)$ is :

[JEE(Main) 2018]

$(x + \sqrt{x^3 - 1})^5 + (x - \sqrt{x^3 - 1})^5$, $(x > 1)$ के प्रसार में सभी विषम घातों वाले पदों के गुणांकों का योग है -

- (A) 1 (B) 2 (C) -1 (D) 0

Ans. B

Sol. $f(x) = (x + \sqrt{x^3 - 1})^5 + (x - \sqrt{x^3 - 1})^5$

$$= 2(x^5 + 10x^3(x^3 - 1) + 5x(x^3 - 1)^2)$$

$$\text{Sum of coefficients of odd degree} = \frac{f(1) - f(-1)}{2} = 2$$

35. For $r = 0, 1, \dots, 10$, let A_r , B_r and C_r denote, respectively, the coefficient of x^r in the expansions of $(1+x)^{10}$,

$(1+x)^{20}$ and $(1+x)^{30}$. Then $\sum_{r=1}^{10} A_r (B_{10} B_r - C_{10} A_r)$ is equal to :

[JEE 2010]

$(1+x)^{10}$, $(1+x)^{20}$ और $(1+x)^{30}$ के प्रसार में माना x^r के गुणांक क्रमशः A_r , B_r और C_r है जहाँ $r = 0, 1, \dots, 10$, तब

$$\sum_{r=1}^{10} A_r (B_{10} B_r - C_{10} A_r) \text{ बराबर:}$$

(A) $B_{10} - C_{10}$

(B) $A_{10} (B_{10}^2 - C_{10} A_{10})$

(C) 0

(D) $C_{10} - B_{10}$

Ans. D

Sol. $A_r = {}^{10}C_r$, $B_r = {}^{20}C_r$, $C_r = {}^{30}C_r$

$$\sum_{r=1}^{10} \left({}^{20}C_{10} {}^{10}C_r {}^{20}C_r - {}^{30}C_{10} ({}^{10}C_r)^2 \right)$$

$$= {}^{20}C_{10} \left({}^{10}C_1 {}^{20}C_1 + {}^{10}C_2 {}^{20}C_2 + \dots + {}^{10}C_{10} {}^{20}C_{10} \right)$$

$$- {}^{30}C_{10} \left({}^{10}C_1^2 + {}^{10}C_2^2 + \dots + {}^{10}C_{10}^2 \right)$$

$$= {}^{20}C_{10} ({}^{30}C_{10} - 1) - {}^{30}C_{10} ({}^{20}C_{10} - 1)$$

$$= {}^{30}C_{10} - {}^{20}C_{10} = C_{10} - B_{10}$$

BINOMIAL THEOREM FOR ANY INDEX

36. Number of terms in the expansion of $(1 - 4x)^{-30}$ are :

(A) 29

(B) 30

(C) 31

(D) infinitely many

$(1 - 4x)^{-30}$ के प्रसार में पदों की संख्या होगी :

(A) 29

(B) 30

(C) 31

; क वृद्ध अनन्त

Ans. D

Sol. No. of terms in the expansion of $(1 - 4x)^{-30}$ are infinite.

Exercise J-II

SUBJECTIVE / INTEGER TYPE QUESTIONS

BASED ON IMPORTANT TERMS

1. (a) If the coefficients of $(2r+4)^{\text{th}}$, $(r-2)^{\text{th}}$ terms in the expansion of $(1+x)^{18}$ are equal, find r .
(b) If the coefficients of the r^{th} , $(r+1)^{\text{th}}$ & $(r+2)^{\text{th}}$ terms in the expansion of $(1+x)^{14}$ are in AP, find r .
(c) If the coefficients of 2^{nd} , 3^{rd} & 4^{th} terms in the expansion of $(1+x)^{2n}$ are in AP, show that $2n^2 - 9n + 7 = 0$.
(a) यदि $(1+x)^{18}$ के प्रसार में $(2r+4)^{\text{th}}$, $(r-2)^{\text{th}}$ पदों के गुणांक समान है तो r का मान ज्ञात कीजिए।
(b) यदि $(1+x)^{14}$ के प्रसार में r^{th} , $(r+1)^{\text{th}}$ तथा $(r+2)^{\text{th}}$ पद समान्तर श्रेणी में है तो r का मान ज्ञात कीजिए।
(c) $(1+x)^{2n}$ के प्रसार में 2^{nd} , 3^{rd} तथा 4^{th} पद समान्तर श्रेणी में है तो सिद्ध कीजिए $2n^2 - 9n + 7 = 0$.

Ans. (a) $r = 6$ (b) $r = 5$ or 9

Sol. (a) Coefficient of T_{2r+4} term = Coefficient of T_{r-2} in $(1+x)^{18}$

$$\Rightarrow {}^{18}C_{2r+3} = {}^{18}C_{r-3}$$

$$\Rightarrow 2r+3 = r-3 \quad \text{or} \quad 2r+r+3-3 = 18$$

$$\Rightarrow r = 6 \quad \text{or} \quad r = 6$$

$$\Rightarrow r = 6$$

$$(b) T_r, T_{r+1}, T_{r+2} \rightarrow \text{A.P. } (1+x)^{14}$$

$$\Rightarrow {}^{14}C_{r-1}, {}^{14}C_r, {}^{14}C_{r+1} \rightarrow \text{A.P.}$$

$$2 \cdot {}^{14}C_r = {}^{14}C_{r-1} + {}^{14}C_{r+1}$$

$$\Rightarrow 4 \cdot {}^{14}C_r = {}^{14}C_r + {}^{14}C_{r-1} + {}^{14}C_{r+1} + {}^{14}C_r$$

$$= {}^{15}C_r + {}^{15}C_{r+1}$$

$$4 \cdot {}^{14}C_r = {}^{16}C_{r+1}$$

$$\Rightarrow 4 \cdot \frac{14}{(14-r)r} = \frac{16}{(15-r)(r+1)} \Rightarrow 4 = \frac{16 \times 15}{(15-r)(r+1)}$$

$$\Rightarrow (r+1)(r-15) = -60 \Rightarrow r^2 - 14r + 45 = 0 \Rightarrow r = 9 \text{ or } 5$$

$$(c) T_2, T_3, T_4 \rightarrow \text{A.P. } (1+x)^{2n}$$

$$\Rightarrow 2 \cdot {}^{2n}C_2 = {}^{2n}C_1 + {}^{2n}C_3$$

$$\Rightarrow 4 \cdot {}^{2n}C_2 = {}^{2n}C_2 + {}^{2n}C_1 + {}^{2n}C_3 + {}^{2n}C_2$$

$$\Rightarrow \frac{4 \cdot \frac{2n}{2}}{\frac{2n-2}{2}} = {}^{2n+1}C_2 + {}^{2n+1}C_3 = {}^{2n+2}C_3$$

$$\Rightarrow \frac{4 \cdot \frac{2n}{2}}{\frac{2n-2}{2}} = \frac{(2n+2)(2n+1) \frac{2n}{3}}{(2n-1) \frac{2n-2}{3}}$$

$$\Rightarrow 12(2n-1) = 4n^2 + 6n + 2$$

$$\Rightarrow 4n^2 + 6n + 2 - 24n + 12 = 0$$

$$\Rightarrow 4n^2 - 18n + 14 = 0$$

$$\Rightarrow 2n^2 - 9n + 7 = 0$$

2. (a) Find the term independent of x in the expansion of (i) $\left[\sqrt{\frac{x}{3}} + \frac{\sqrt{3}}{2x^2}\right]^{10}$ (ii) $\left[\frac{1}{2}x^{1/3} + x^{-1/5}\right]^8$
- (b) Find the value of x for which the fourth term in the expansion, $\left(5^{\frac{2}{5}\log_5 \sqrt{4^x+44}} + \frac{1}{5^{\log_5 \sqrt[3]{2^{x-1}+7}}}\right)^8$ is 336.
- (a) निम्न के प्रसार में x रहित पद ज्ञात कीजिए। (i) $\left[\sqrt{\frac{x}{3}} + \frac{\sqrt{3}}{2x^2}\right]^{10}$ (ii) $\left[\frac{1}{2}x^{1/3} + x^{-1/5}\right]^8$
- (b) x का मान ज्ञात कीजिए जिसके लिए $\left(5^{\frac{2}{5}\log_5 \sqrt{4^x+44}} + \frac{1}{5^{\log_5 \sqrt[3]{2^{x-1}+7}}}\right)^8$ के प्रसार में 4th पद 336 है।

Ans. (a) (i) $\frac{5}{12}$ (ii) $T_6 = 7$, (b) $x = 0$ or 1

Sol. (a) (i) $\left[\sqrt{\frac{x}{3}} + \frac{\sqrt{3}}{2x^2}\right]^{10}$

$$= \sum_{r=0}^{10} {}^{10}C_r \left(\frac{x}{3}\right)^{\frac{10-r}{2}} \left(\frac{\sqrt{3}}{2x^2}\right)^r$$

Term independent of x if

$$5 - \frac{r}{2} - 2r = 0 \Rightarrow 10 - 5r = 2$$

$$\Rightarrow r = 2$$

$$\text{Term} = {}^{10}C_2 \cdot \left(\frac{1}{3}\right)^4 \cdot \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{45}{27} \cdot \frac{1}{4} = \frac{5}{12}$$

$$(ii) \left[\frac{1}{2} x^{\frac{1}{3}} + x^{-\frac{1}{5}} \right]^8 = \sum_{r=0}^8 {}^8C_r \left(\frac{1}{2}\right)^{8-r} (x)^{\frac{8-r}{3}} \cdot x^{-\frac{r}{5}}$$

$$\text{Term independent of x if } \frac{8-r}{3} - \frac{r}{5} = 0$$

$$\Rightarrow 40 - 5r - 3r = 0$$

$$\Rightarrow r = 5$$

$$\text{Term } T_6 = {}^8C_5 \left(\frac{1}{2}\right)^3 = \frac{8}{5 \cdot 3} \cdot \frac{1}{8} = 7$$

$$(b) \left(5^{\frac{2}{5} \log_5 \sqrt{4^x + 44}} + \frac{1}{5^{\log_5 \sqrt[3]{2^{x-1} + 7}}} \right)^8$$

$$= \left(\left(4^x + 44 \right)^{\frac{2}{5} \cdot 2} + \frac{1}{\left(2^{x-1} + 7 \right)^{\frac{1}{3}}} \right)^8$$

$$\text{Given } T_4 = 336$$

$$\Rightarrow {}^8C_3 (4^x + 44)^{\frac{8-3}{5}} \cdot \left(\frac{1}{(2^{x-1} + 7)^{\frac{1}{3}}} \right)^3 = 336$$

$$\Rightarrow \frac{4^x + 44}{2^{x-1} + 7} = 6$$

$$\text{Let } 2^x = t$$

$$\Rightarrow t^2 + 44 = 3t + 42$$

$$\Rightarrow t^2 - 3t + 2 = 0$$

$$\Rightarrow t = 1 \text{ or } t = 2$$

$$\Rightarrow x = 0 \text{ or } x = 1$$

BASED ON FINDING COEFFICIENTS

3. Prove that the ratio of the coefficient of x^{10} in $(1-x^2)^{10}$ & the term independent of x in $\left(x - \frac{2}{x}\right)^{10}$ is 1 : 32.

सिद्ध कीजिए $(1-x^2)^{10}$ के प्रसार में x^{10} का गुणांक तथा $\left(x - \frac{2}{x}\right)^{10}$ के प्रसार में x रहित पद के गुणांक का अनुपात 1 : 32 है।

Ans.

Sol. $(1-x^2)^{10} = \sum_{r=0}^{10} {}^{10}C_r (-x^2)^r$

Coefficient of x^{10} is ${}^{10}C_5 (-1)^5$

$$\left(x - \frac{2}{x}\right)^{10} = \sum_{r=0}^{10} {}^{10}C_r x^{10-r} \cdot (-2)^r \cdot x^{-r}$$

Term independent of x if $r = 5$

$$\Rightarrow \text{Coefficient of term independent of } x = {}^{10}C_5 (-2)^5$$

$$\Rightarrow \frac{{}^{10}C_5 (-1)^5}{{}^{10}C_5 (-2)^5} = \frac{1}{32}$$

4. Let a and b be the coefficient of x^3 in $(1+x+2x^2+3x^3)^4$ and $(1+x+2x^2+3x^3+4x^4)^4$ respectively. Find the value of $(a-b)$.

यदि $(1 + x + 2x^2 + 3x^3)^4$ तथा $(1 + x + 2x^2 + 3x^3 + 4x^4)^4$ के प्रसार में x^3 के गुणांक क्रमशः a तथा b है तो $(a - b)$ का मान होगा ।

Ans. 0

Sol. Coefficient of x^3 in $(1 + x + 2x^2 + 3x^3)^4$

$$= \frac{|4|}{|P_1|P_2|P_3|P_4|} \cdot 1^{P_1} \cdot x^{P_2} \cdot 2^{P_3} \cdot x^{2P_3} \cdot 3^{P_4} \cdot x^{3P_4}$$

$$= \frac{|4|}{|P_1|P_2|P_3|P_4|} \cdot 1^{P_1} \cdot 2^{P_3} \cdot 3^{P_4} \cdot x^{P_2+2P_3+3P_4}$$

$$P_1 + P_2 + P_3 + P_4 = 4$$

$$P_2 + 2P_3 + 3P_4 = 3$$

$$\Rightarrow a = 40$$

$$(1 + x + 2x^2 + 3x^3 + 4x^4)^4 = \frac{|4| \cdot 2^{P_3} \cdot 3^{P_4} \cdot 4^{P_5}}{|P_1|P_2|P_3|P_4|P_5|} x^{P_2+2P_3+3P_4+4P_5}$$

For x^3

$$P_1 + P_2 + P_3 + P_4 + P_5 = 4$$

$$P_2 + 2P_3 + 3P_4 + 4P_5 = 3$$

$$\Rightarrow b = 40$$

$$\Rightarrow b - a = 0$$

5. Find the coefficient of x^r in the expression of

$$(x + 3)^{n-1} + (x + 3)^{n-2}(x + 2) + (x + 3)^{n-3}(x + 2)^2 + \dots + (x + 2)^{n-1}.$$

$(x+3)^{n-1} + (x+3)^{n-2}(x+2) + (x+3)^{n-3}(x+2)^2 + \dots + (x+2)^{n-1}$ में x^r का गुणांक ज्ञात कीजिए।

Ans. ${}^nC_r(3^{n-r} - 2^{n-r})$

Sol. $(x+3)^{n-1} + (x+3)^{n-2}(x+2) + \dots + (x+2)^{n-1}(x+3)^{n-1}$

G.P. with $r = \frac{(x+2)}{x+3}$

$$\begin{aligned} &= (x+3)^{n-1} \frac{\left[\left(\frac{x+2}{x+3} \right)^n - 1 \right]}{\frac{x+2}{x+3} - 1} \\ &= (x+3)^{n-1} \cdot \frac{(x+3)}{-1} \left[\left(\frac{x+2}{x+3} \right)^n - 1 \right] \\ &= -(x+3)^n \left[\left(\frac{x+2}{x+3} \right)^n - 1 \right] \\ &= (x+3)^n - (x+2)^n \end{aligned}$$

Coefficient of $x^r = {}^nC_r(3^{n-r} - 2^{n-r})$

6. Find the term independent of x in the expansion of $(1+x+2x^3) \left(\frac{3x^2}{2} - \frac{1}{3x} \right)^9$.

$(1+x+2x^3) \left(\frac{3x^2}{2} - \frac{1}{3x} \right)^9$ के प्रसार में x रहित पद ज्ञात कीजिए।

Ans. $\frac{17}{54}$

Sol. Co-efficient of x^0 in $(1+x+2x^3) \left(\frac{3}{2}x^2 - \frac{1}{3x} \right)^9$
 $=$ Co-efficient of x^0 in $\left(\frac{3}{2}x^2 - \frac{1}{3x} \right)^9$ + Co-efficient of x^{-1} in $\left(\frac{3}{2}x^2 - \frac{1}{3x} \right)^9$
 $+ 2$ Co-efficient of x^{-3} in $\left(\frac{3}{2}x^2 - \frac{1}{3x} \right)^9$
 $= {}^9C_{r_1} \left(\frac{3}{2}x^2 \right)^{9-r_1} \left(-\frac{1}{3x} \right)^{r_1} + {}^9C_{r_2} \left(\frac{3}{2}x^2 \right)^{9-r_2} \left(-\frac{1}{3x} \right)^{r_2} + 2 {}^9C_{r_3} \left(\frac{3}{2}x^2 \right)^{9-r_3} \left(-\frac{1}{3x} \right)^{r_3}$

$$= {}^9C_6 \left(\frac{3}{2}\right)^3 \left(-\frac{1}{3}\right)^6 + 2 {}^9C_7 \left(\frac{3}{2}\right)^2 \left(-\frac{1}{3}\right)^7 = \frac{17}{54}$$

7. Let $(1+x^2)^2 \cdot (1+x)^n = \sum_{k=0}^{n+4} a_k \cdot x^k$. If a_1, a_2 & a_3 are in AP, find n .

माना $(1+x^2)^2 \cdot (1+x)^n = \sum_{k=0}^{n+4} a_k \cdot x^k$. यदि a_1, a_2 तथा a_3 समान्तर श्रेणी में है तो n का मान ज्ञात कीजिए।

Ans. $n = 2$ or 3 or 4

Sol.
$$(1+x^2)^2 (1+x)^n = \sum_{k=0}^{n+4} a_k \cdot x^k$$

$$\Rightarrow (1+2x^2+x^4)(1+x)^n = \sum_{k=0}^{n+4} a_k \cdot x^k$$

$$\Rightarrow (1+2x^2+x^4) \sum_{r=0}^n {}^nC_r x^r = a_0 + a_1 x + \dots$$

$$a_1 = {}^nC_1 = n$$

$$a_2 = {}^nC_2 + 2 {}^nC_0 = \frac{n(n-1)}{2} + 2 = \frac{n^2 - n + 4}{2}$$

$$a_3 = {}^nC_3 + 2 {}^nC_1 = \frac{n(n-1)(n-2)}{6} + 2n$$

$$\Rightarrow a_1 = n, a_2 = \frac{n^2 - n + 4}{2}, a_3 = \frac{n(n-1)(n-2)}{6} + 2n$$

Given $2a_2 = a_1 + a_3$

$$\Rightarrow n^2 - n + 4 = n + \frac{n(n-1)(n-2)}{6} + 2n$$

$$\Rightarrow n^2 - 2n + 4 = \frac{(n^2 - n)(n-2)}{6} + 2n$$

$$\Rightarrow n^2 - 2n + 4 - 2n = \frac{n(n-1)(n-2)}{6}$$

$$\Rightarrow n(n-2) + (-2)(n-2) = \frac{n(n-1)(n-2)}{6}$$

$$\Rightarrow 6(n-2)^2 - (n-2)n(n-1) = 0$$

$$\Rightarrow (n-2)[6n-12-n^2+n] = 0$$

$$(n-2)[7n-12-n^2] = 0$$

$$n = 2, 3, 4$$

8. Find the sum of the roots (real or complex) of the equation $x^{2001} + \left(\frac{1}{2} - x\right)^{2001} = 0$.

समीकरण $x^{2001} + \left(\frac{1}{2} - x\right)^{2001} = 0$ के सभी मूलों (वास्तविक या काल्पनिक) का योगफल ज्ञात कीजिए।

Ans. 500

Sol. $x^{2001} + \left(\frac{1}{2} - x\right)^{2001} = 0$

$$\therefore \left(\frac{1}{2} - x\right)^{2001} = \sum_{r=0}^{2001} {}^{2001}C_r \left(\frac{1}{2}\right)^{2001-r} \cdot (-x)^r$$

$$= {}^{2001}C_0 \left(\frac{1}{2}\right)^{2001} + {}^{2001}C_1 \left(\frac{1}{2}\right)^{2000} (-x) + {}^{2001}C_2 \left(\frac{1}{2}\right)^{2000} (-x)^2 + \dots + {}^{2001}C_{2000} \left(\frac{1}{2}\right)^1 (-x)^{2000} + {}^{2001}C_{2001} (-x)^{2001}$$

$$\Rightarrow \sum_r {}^{2001}C_r \left(\frac{1}{2}\right)^{2001-r} (-x)^r = 0$$

Sum of the roots of this equation

$$= - \frac{\text{Coefficient of } x^{1999}}{\text{coefficient of } x^{2000}}$$

$$= + \frac{\left(\frac{1}{2}\right)^2 \cdot {}^{2001}C_{1999}}{\left(\frac{1}{2}\right)^1 \cdot {}^{2001}C_{2000}}$$

$$= \frac{1}{4} \times \frac{2}{1} \times \frac{|2001|!|2000|}{|2|1999|2001|}$$

$$= \frac{1}{4} \times 2000 = 500$$

NUMERICALLY GREATEST TERM

9. Find numerically greatest term in the expansion of

(i) $(2 + 3x)^9$ when $x = \frac{3}{2}$ (ii) $(3 - 5x)^{15}$ when $x = \frac{1}{5}$

निम्न के प्रसार में संख्यात्मक महत्तम पद होगा

(i) $(2 + 3x)^9$ जब $x = \frac{3}{2}$ (ii) $(3 - 5x)^{15}$ जब $x = \frac{1}{5}$

Ans. (i) $T_7 = \frac{7 \cdot 3^{13}}{2}$ (ii) 455×3^{12}

Sol. (i) $(2 + 3x)^9$; $x = \frac{3}{2}$

$$p = \frac{n+1}{1 + \left| \frac{2}{3x} \right|} = \frac{10}{1 + \left| \frac{2}{3 \times \frac{3}{2}} \right|} = \frac{10}{1 + \frac{4}{9}} = \frac{90}{13}$$

$$p \notin \mathbb{Z}$$

$$\Rightarrow \text{NGT is } T_{[p]+1} = T_7 = {}^9C_6 (2)^3 \cdot 3^6$$

$$= \frac{9 \times 8 \times 7}{6} \times 8 \times 3^6 \times \left(\frac{3}{2}\right)^6$$

$$= \frac{3^{13} \cdot 7}{2}$$

(ii) $(3 - 5x)^{15}$ when $x = \frac{1}{5}$

$$p = \frac{15+1}{1 + \left| \frac{3}{-5x} \right|} = \frac{16}{1+3} = 4 \in \mathbb{Z}$$

$$\Rightarrow T_4 \text{ \& } T_{4+1} = T_5 \text{ are NGT}$$

$$T_5 = {}^{15}C_4 (3)^{11} (-5x)^4 \Big|_{x=\frac{1}{5}}$$

$$= \frac{15 \times 14 \times 13 \times 12}{4 \times 3 \times 2} \times 3^{11}$$

$$= 3^{12} \times 455 = T_5$$

$$T_4 = {}^{15}C_3 3^{12} \cdot (-5x)^4$$

$$\Rightarrow T_4 = \frac{15 \times 14 \times 13}{3 \times 2} \times -3^{12}$$

$$\Rightarrow T_4 = -455 \times 3^{12}$$

$$\Rightarrow |T_4| = |T_5|$$

10. Find the index n of the binomial $\left(\frac{x}{5} + \frac{2}{5}\right)^n$ if the 9th term of the expansion has numerically the greatest coefficient ($n \in \mathbb{N}$).

n का मान ज्ञात कीजिए जिसके लिए $\left(\frac{x}{5} + \frac{2}{5}\right)^n$ के प्रसार में 9th पद संख्यात्मक महत्तम है ($n \in \mathbb{N}$).

Ans. $n = 12$

Sol. For T_9 to be the numerically greatest term, $r = \left[\frac{n+1}{1 + \left| \frac{x}{a} \right|} \right] = \left[\frac{n+1}{1 + \left| \frac{1}{2} \right|} \right] = 8$

$$\Rightarrow 8 < \frac{2(n+1)}{3} < 9 \Rightarrow 11 < n < 12.5 \Rightarrow n = 12$$

11. For which positive values of x is the fourth term in the expansion of $(5 + 3x)^{10}$ is the greatest.

x के किस मान के लिए $(5 + 3x)^{10}$ के प्रसार में 4th पद संख्यात्मक महत्तम है।

Ans. $\frac{5}{8} < x < \frac{20}{21}$

Sol. 4th term is greatest in the expansion of $(5 + 3x)^{10}$

$$\Rightarrow [P] = 3$$

$$\Rightarrow 3 < P < 4$$

$$\Rightarrow 3 < \frac{10+1}{1 + \left| \frac{5}{3x} \right|} < 4$$

$$\Rightarrow 3 + 3\left|\frac{5}{3x}\right| < 11 \quad \& \quad 11 < 4 + 4\left|\frac{5}{3x}\right|$$

$$\frac{15}{3|x|} < 8 \quad \& \quad \frac{7}{20} < \frac{1}{3|x|}$$

$$\Rightarrow 8|x| > 5 \quad \& \quad |x| < \frac{20}{21}$$

$$\Rightarrow |x| > \frac{5}{8} \quad \& \quad |x| < \frac{20}{21}$$

$$\Rightarrow x \in \left(-\infty, -\frac{5}{8}\right) \cup \left(\frac{5}{8}, \infty\right) \quad \& \quad x \in \left(-\frac{20}{21}, \frac{20}{21}\right)$$

$$\Rightarrow x \in \left(-\frac{20}{21}, -\frac{5}{8}\right) \cup \left(\frac{5}{8}, \frac{20}{21}\right)$$

Given $x > 0$

$$\Rightarrow x \in \left(\frac{5}{8}, \frac{20}{21}\right)$$

MULTINOMIAL THEOREM

12. Find the coefficient of

(a) $x^2 y^3 z^4$ in the expansion of $(ax - by + cz)^9$.

(b) $a^2 b^3 c^4 d$ in the expansion of $(a - b - c + d)^{10}$.

निम्न के गुणांक ज्ञात कीजिए।

(a) $(ax - by + cz)^9$ के प्रसार में $x^2 y^3 z^4$ का गुणांक

(b) $(a - b - c + d)^{10}$ के प्रसार में $a^2 b^3 c^4 d$ का गुणांक

Ans. (a) $-1260 \cdot a^2 b^3 c^4$; (b) -12600

Sol. (a) $(ax - by + cz)^9 = \sum_{P_1+P_2+P_3=9} \frac{9!}{P_1!P_2!P_3!} a^{P_1} (-b)^{P_2} (+c)^{P_3} x^{P_1} y^{P_2} z^{P_3}$

For $x^2 y^3 z^4$

$$P_1 = 2, P_2 = 3, P_3 = 4$$

$$\Rightarrow \text{Coefficient} = -\frac{|9|}{|2|3|4|} a^2 b^3 c^4$$

$$= -\frac{9 \times 8 \times 7 \times 6 \times 5}{2 \times 6} a^2 b^3 c^4$$

$$= -63 \times 20 a^2 b^3 c^4$$

$$= -1260 a^2 b^3 c^4$$

$$(b) (a - b - c + d)^{10}$$

$$= \sum_{P_1+P_2+P_3+P_4=10} \frac{|10| a^{P_1} (-b)^{P_2} (-c)^{P_3} d^{P_4}}{|P_1|P_2|P_3|P_4|}$$

For $a^2 b^3 c^4 d$

$$P_1 = 2, P_2 = 3, P_3 = 4, P_4 = 1$$

Coefficient of $a^2 b^3 c^4 d$

$$= \frac{|10|}{|2|3|4|1|} (-1)^3 (-1)^4$$

$$= -\frac{10 \times 9 \times 8 \times 7 \times 6 \times 5}{2 \times 6} = -20 \times 63 \times 10 = -12600$$

13. Find the coefficient of

$$(a) x^4 \text{ in the expansion of } (1 + x + x^2 + x^3)^{11} \quad (b) x^4 \text{ in the expansion of } (2 - x + 3x^2)^6.$$

निम्न के गुणांक ज्ञात कीजिए।

$$(a) (1 + x + x^2 + x^3)^{11} \text{ के प्रसार में } x^4 \text{ का गुणांक}$$

$$(b) (2 - x + 3x^2)^6 \text{ के प्रसार में } x^4 \text{ का गुणांक}$$

Ans. (a) 990 (b) 3660

$$\text{Sol. } (1 + x + x^2 + x^3)^{11} = \sum_{P_1+P_2+P_3+P_4=11} \frac{|11| \cdot x}{|P_1|P_2|P_3|P_4|}$$

For x^4

$$P_4 + 2P_3 + 3P_2 = 4$$

$$(b) \left(2 - x + 3x^2\right)^6 = \sum_{r_1+r_2+r_3=6} \frac{62^{r_1} \cdot (-1)^{r_2} (3)^{r_3}}{\underline{r_1} \underline{r_2} \underline{r_3}} x^{r_2+2r_3}$$

For x^4

$$r_1 + r_2 + r_3 = 6 \text{ \& } r_2 + 2r_3 = 4$$

INTEGRAL PART AND FRACTIONAL PART BASED CONCEPT

14. (a) Show that the integral part in each of the following is odd. $n \in \mathbb{N}$

$$(A) \left(5 + 2\sqrt{6}\right)^n \quad (B) \left(8 + 3\sqrt{7}\right)^n$$

(b) Show that the integral part in each of the following is even. $n \in \mathbb{N}$

$$(A) \left(3\sqrt{3} + 5\right)^{2n+1} \quad (B) \left(5\sqrt{5} + 11\right)^{2n+1}$$

(a) सिद्ध कीजिए प्रत्येक का पूर्णांक भाग विषम है जहाँ $n \in \mathbb{N}$

$$(A) \left(5 + 2\sqrt{6}\right)^n \quad (B) \left(8 + 3\sqrt{7}\right)^n$$

(b) सिद्ध कीजिए प्रत्येक का पूर्णांक भाग सम है जहाँ $n \in \mathbb{N}$

$$(A) \left(3\sqrt{3} + 5\right)^{2n+1} \quad (B) \left(5\sqrt{5} + 11\right)^{2n+1}$$

Ans.

Sol. (a) (A) $(5 + 2\sqrt{6})^n = I + f; 0 < f < 1$

$$(5 - 2\sqrt{6})^n = 0 + f'; 0 < f' < 1$$

$$\Rightarrow I + f + f' = (5 + 2\sqrt{6})^n + (5 - 2\sqrt{6})^n$$

$$= 2 \left[{}^nC_0 5^n + {}^nC_2 5^{n-2} (2\sqrt{6})^2 + \dots \right]$$

$$\Rightarrow I + f + f' = \text{even Integer}$$

$$\therefore 0 < f < 1 \text{ \& } 0 < f' < 1$$

$$\Rightarrow 0 < f + f' < 2$$

Case-I : If I is even integer then $f + f'$ must be even integer.

but $0 < f + f' < 2 \rightarrow$ There is no even integer

Case-II : If I is odd integer then $f + f'$ must be odd integer

$$\Rightarrow f + f' = \text{as } 0 < f + f' < 2$$

$\therefore$ Integral part of $(5 + 2\sqrt{6})^n$ is odd.

$$(B) (8 + 3\sqrt{7})^n = I + f; 0 < f < 1$$

$$(8 - 3\sqrt{7})^n = 0 + f'; 0 < f' < 1$$

$$\Rightarrow (8 + 3\sqrt{7})^n + (8 - 3\sqrt{7})^n = 2 \left[{}^nC_0 8^n + {}^nC_2 8^{n-2} (3\sqrt{7})^2 + \dots \right] = I + f + f'$$

$$\Rightarrow I + f + f' = \text{even integer}$$

$$\therefore 0 < f < 1 \quad \& \quad 0 < f' < 1$$

$$\Rightarrow 0 < f + f' < 2 \Rightarrow f + f' = 1$$

$\Rightarrow$ Integral part of $(8 + 3\sqrt{7})^n$ is odd.

$$(b) (A) (3\sqrt{3} + 5)^{2n+1} = I + f; 0 < f < 1$$

$$(3\sqrt{3} - 5)^{2n+1} = f'; 0 < f' < 1$$

$\downarrow$
5.19

$$\begin{aligned} \Rightarrow I - f - f' &= (3\sqrt{3} + 5)^{2n+1} - (3\sqrt{3} - 5)^{2n+1} \\ &= 2 \left[{}^{2n+1}C_1 (3\sqrt{3})^{2n} (5)^1 + \dots \right] \end{aligned}$$

$$\Rightarrow I + f - f' = \text{even integer}$$

$$-1 < -f' < 0$$

$$-1 < f - f' < 1$$

$$\Rightarrow f - f' = 0$$

$$\Rightarrow I \text{ must be even}$$

$$(B) (5\sqrt{5} + 11)^{2n+1} = I + f$$

$$(5\sqrt{5} - 11)^{2n+1} = f'$$

$$\begin{aligned} \Rightarrow I + f - f' &= (5\sqrt{5} + 11)^{2n+1} - (5\sqrt{5} - 11)^{2n+1} \\ &= 2 \left[{}^{2n+1}C_1 (5\sqrt{5})^{2n} \cdot 1 + \dots \right] \end{aligned}$$

$$\therefore -1 < f - f' < 0$$

$$\Rightarrow f - f' = 0$$

$$\Rightarrow I \text{ must be even}$$

15. If $(7 + 4\sqrt{3})^n = p + \beta$ where n & p are positive integers and β is a proper fraction show that

$$(1 - \beta)(p + \beta) = 1.$$

यदि $(7 + 4\sqrt{3})^n = p + \beta$ जहाँ n & p धनात्मक पूर्णांक हैं तथा β भिन्नात्मक भाग है तो सिद्ध कीजिए

$$(1 - \beta)(p + \beta) = 1.$$

Ans.

Sol. $(7 + 4\sqrt{3})^n = p + \beta$; $n \in \mathbb{N}$ & $p \in \mathbb{N}$ & β is a proper fraction.

$$(7 - 4\sqrt{3})^n = \beta'; 0 < \beta' < 1$$

$$\Rightarrow \beta' = \frac{(49 - 48)^n}{(7 + 4\sqrt{3})^n} = \frac{1}{p + \beta}$$

$$\Rightarrow p + \beta + \beta' = (7 + 4\sqrt{3})^n + (7 - 4\sqrt{3})^n$$

$$= 2 \left[{}^nC_0 7^n + {}^nC_2 7^{n-2} (4\sqrt{3})^2 + \dots \right]$$

$$\Rightarrow \beta + \beta' < 2$$

$$\Rightarrow \beta + \beta' = 1$$

$$\Rightarrow 1 - \beta = \beta'$$

$$\Rightarrow 1 - \beta = \frac{1}{p + \beta}$$

$$\Rightarrow (p + \beta)(1 - \beta) = 1$$

16. If $(6\sqrt{6} + 14)^{2n+1} = N$ & F be the fractional part of N , prove that $NF = 20^{2n+1}$ ($n \in \mathbb{N}$)

यदि $(6\sqrt{6} + 14)^{2n+1} = N$ तथा F , N का भिन्नात्मक भाग है तो सिद्ध कीजिए $NF = 20^{2n+1}$ ($n \in \mathbb{N}$)

Ans.

Sol. $(6\sqrt{6} + 14)^{2n+1} = I + f = N$; $0 < f < 1$

$$(6\sqrt{6} - 14)^{2n+1} = f'$$

$$\Rightarrow I + f - f' = 2 \left({}^{2n+1}C_1 (6\sqrt{6})^{2n} + {}^{2n+1}C_3 (6\sqrt{6})^{2n-2} + \dots \right)$$

$$\Rightarrow f - f' = 0$$

$$\Rightarrow f = f'$$

$$\Rightarrow Nf = Nf' = (6\sqrt{6} + 14)^{2n+1} \cdot (6\sqrt{6} - 14)^{2n+1}$$

$$= (216 - 196)^{2n+1}$$

$$= 20^{2n+1}; n \in \mathbb{N}$$

17. Let I denotes the integral part and F the proper fractional part of $(3 + \sqrt{5})^n$ where $n \in \mathbb{N}$ and if ρ denotes the rational part and σ the irrational part of the same, show that

$$\rho = \frac{1}{2}(I + 1) \text{ and } \sigma = \frac{1}{2}(I + 2F - 1).$$

यदि I तथा F , $(3 + \sqrt{5})^n$ के प्रसार में क्रमशः पूर्णांक तथा भिन्नात्मक भाग हैं जहाँ $n \in \mathbb{N}$ तथा यदि ρ परिमेय भाग तथा σ अपरिमेय भाग है तो सिद्ध कीजिए

$$\rho = \frac{1}{2}(I + 1) \text{ and } \sigma = \frac{1}{2}(I + 2F - 1).$$

Ans.

Sol. $(3 + \sqrt{5})^n = I + F$

$$(3 - \sqrt{5})^n = F'$$

$$\Rightarrow I + F + F' = 2 \left[{}^nC_0 3^n + {}^nC_2 3^{n-2} (\sqrt{5})^2 + \dots \right]$$

$$\Rightarrow 0 < F + F' < 2$$

$$\Rightarrow F + F' = 1$$

$$\Rightarrow F' = 1 - F$$

$$\therefore (I+F) = (3+\sqrt{5})^n = \left[{}^nC_0 3^n (\sqrt{5})^0 + {}^nC_1 3^{n-1} (\sqrt{5})^1 + {}^nC_2 3^{n-2} (\sqrt{5})^2 + \dots \right]$$

$$(F') = (3-\sqrt{5})^n = \left[{}^nC_0 3^n (\sqrt{5})^0 - {}^nC_1 3^{n-1} (\sqrt{5})^1 + \dots \right]$$

$$\Rightarrow 2\rho = I + F + F' \Rightarrow \rho = \frac{I+1}{2}$$

$$\& 2\sigma = I + F - F' \Rightarrow \sigma = \frac{I+2F-1}{2}$$

SUMMATION OF SERIES

18. Given $s_n = 1 + q + q^2 + \dots + q^n$ & $S_n = 1 + \frac{q+1}{2} + \left(\frac{q+1}{2}\right)^2 + \dots + \left(\frac{q+1}{2}\right)^n$, $q \neq 1$,
prove that ${}^{n+1}C_1 + {}^{n+1}C_2 \cdot s_1 + {}^{n+1}C_3 \cdot s_2 + \dots + {}^{n+1}C_{n+1} \cdot s_n = 2^n \cdot S_n$.

दिया गया है $s_n = 1 + q + q^2 + \dots + q^n$ और $S_n = 1 + \frac{q+1}{2} + \left(\frac{q+1}{2}\right)^2 + \dots + \left(\frac{q+1}{2}\right)^n$, $q \neq 1$,
तो सिद्ध कीजिए ${}^{n+1}C_1 + {}^{n+1}C_2 \cdot s_1 + {}^{n+1}C_3 \cdot s_2 + \dots + {}^{n+1}C_{n+1} \cdot s_n = 2^n \cdot S_n$.

Ans.

Sol. $s_n = \frac{q^{n+1}-1}{q-1}$ and $S_n = \frac{\left(\frac{q+1}{2}\right)^{n+1}-1}{\frac{q+1}{2}-1}$

$${}^{n+1}C_1 + {}^{n+1}C_2 s_1 + \dots + {}^{n+1}C_{n+1} s_n$$

$$= {}^{n+1}C_1 + {}^{n+1}C_2 \left(\frac{q^2-1}{q-1}\right) + \dots + {}^{n+1}C_{n+1} \frac{q^{n+1}-1}{q-1}$$

$$= \frac{1}{q-1} [{}^{n+1}C_1 q + {}^{n+1}C_2 q^2 + \dots + {}^{n+1}C_{n+1} q^{n+1} - {}^{n+1}C_1 - {}^{n+1}C_2 - \dots - {}^{n+1}C_{n+1}]$$

$$= \frac{1}{q-1} [(1+q)^{n+1} - 1 - 2^{n+1} + 1] = \frac{1}{q-1} [(1+q)^{n+1} - 2^{n+1}] = \left(\frac{\left(\frac{q+1}{2}\right)^{n+1} - 1}{\frac{q-1}{2}} \right) \cdot 2^n = 2^n S_n$$

19. Find the sum of the series $\sum_{r=0}^n (-1)^r \cdot {}^nC_r \left[\frac{1}{2^r} + \frac{3^r}{2^{2r}} + \frac{7^r}{2^{3r}} + \frac{15^r}{2^{4r}} + \dots \text{up to } m \text{ terms} \right]$

श्रेणी $\sum_{r=0}^n (-1)^r \cdot {}^nC_r \left[\frac{1}{2^r} + \frac{3^r}{2^{2r}} + \frac{7^r}{2^{3r}} + \frac{15^r}{2^{4r}} + \dots \text{up to } m \text{ terms} \right]$ का योगफल ज्ञात कीजिए।

Ans. $\frac{(2^{mn} - 1)}{(2^n - 1)(2^{mn})}$

Sol.
$$\sum_{r=0}^n (-1)^r \cdot {}^nC_r \left[\frac{1}{2^r} + \left(\frac{3}{4}\right)^r + \left(\frac{7}{8}\right)^r + \dots \text{m terms} \right]$$

$$= \sum_{r=0}^n \left[(-1)^r {}^nC_r \frac{1}{2^r} + (-1)^r {}^nC_r \left(\frac{3}{4}\right)^r + \dots \text{m terms} \right]$$

$$= [{}^nC_0 - {}^nC_1 \left(\frac{1}{2}\right) + {}^nC_2 \left(\frac{1}{2}\right)^2 + \dots] +$$

$$[{}^nC_0 - {}^nC_1 \left(\frac{3}{4}\right) + {}^nC_2 \left(\frac{3}{4}\right)^2 + \dots] + \dots \text{m terms}$$

$$= \left(1 - \frac{1}{2}\right)^n + \left(1 - \frac{3}{4}\right)^n + \left(1 - \frac{7}{8}\right)^n + \dots \text{m terms}$$

$$= \frac{1}{2^n} \left(\frac{1 - \frac{1}{2^{mn}}}{1 - \frac{1}{2^n}} \right) = \frac{1}{2^n - 1} \left(1 - \frac{1}{2^{mn}} \right)$$

20. Using the theory of permutation or otherwise prove that

$${}^{100}C_{10} + 5 \cdot {}^{100}C_{11} + 10 \cdot {}^{100}C_{12} + 10 \cdot {}^{100}C_{13} + 5 \cdot {}^{100}C_{14} + {}^{100}C_{15} = {}^{105}C_{90}$$

क्रमचय सिद्धान्त या अन्यथा प्रयोग कर सिद्ध कीजिये।

$${}^{100}C_{10} + 5 \cdot {}^{100}C_{11} + 10 \cdot {}^{100}C_{12} + 10 \cdot {}^{100}C_{13} + 5 \cdot {}^{100}C_{14} + {}^{100}C_{15} = {}^{105}C_{90}.$$

Ans.

Sol.
$${}^{100}C_{10} + 5 \cdot {}^{100}C_{11} + 10 \cdot {}^{100}C_{12} + 10 \cdot {}^{100}C_{13} + 5 \cdot {}^{100}C_{14} + {}^{100}C_{15}$$

$$= {}^5C_0 \cdot {}^{100}C_{10} + {}^5C_1 \cdot {}^{100}C_{11} + {}^5C_2 \cdot {}^{100}C_{12} + {}^5C_3 \cdot {}^{100}C_{13} + {}^5C_4 \cdot {}^{100}C_{14} + {}^5C_5 \cdot {}^{100}C_{15}$$

$$|r_1 - r_2| = -10 = \text{Constant}$$

$$(x + 1)^5 = {}^5C_0 x^5 + {}^5C_1 x^4 + {}^5C_2 x^3 + {}^5C_3 x^2 + {}^5C_4 x^1 + {}^5C_5$$

$$(1 + x)^{100} = {}^{100}C_0 + {}^{100}C_1 x + \dots + {}^{100}C_{15} x^{15} + \dots + {}^{100}C_{100} x^{100}$$

$$(1 + x)^{105} = \left[\sum_{r=0}^5 {}^5C_r x^r \right] \left[\sum_{r=0}^{100} {}^{100}C_r x^r \right]$$

Comparing coefficient of x^{15}

$$^{100}C_{10} + 5 \cdot ^{100}C_{11} + 10 \cdot ^{100}C_{12} + 100 \cdot ^{100}C_{13} + 5 \cdot ^{100}C_{14} + ^{100}C_{15}$$

$$= ^{105}C_{90}$$

21. Let $f(n) = \sum_{r=0}^n \sum_{k=r}^n \binom{k}{r}$. Find the total number of divisors of $f(11)$.

माना $f(n) = \sum_{r=0}^n \sum_{k=r}^n \binom{k}{r}$. तब $f(11)$ के कुल विभाजक ज्ञात कीजिये।

Ans. 24

Sol.
$$f(n) = \sum_{r=0}^n \sum_{k=r}^n \binom{k}{r} = \sum_{r=0}^n \left[\sum_{k=r}^n {}^k C_r \right]$$

$$= \sum_{r=0}^n \left[{}^r C_r + {}^{r+1} C_r + \dots + {}^n C_r \right]$$

$$\sum_{r=0}^n \left[{}^{r+1} C_{r+1} + {}^{r+1} C_r + {}^{r+2} C_r + \dots + {}^n C_r \right]$$

$$= \sum_{r=0}^n \left[{}^{r+2} C_{r+1} + {}^{r+2} C_r + \dots + {}^{n-1} C_r + {}^n C_r \right]$$

$$= \sum_{r=0}^n {}^{n+1} C_{r+1} = {}^{n+1} C_1 + {}^{n+1} C_2 + \dots + {}^{n+1} C_{n+1}$$

$$= 2^{n+1} - 1$$

$$\Rightarrow f(11) = 2^{12} - 1 = 4095 = 5 \times 9 \times 7 \times 13 = 5 \times 7 \times 13 \times 3^2$$

$$\Rightarrow \text{Total no. of divisor} = 2 \times 2 \times 2 \times 3 = 24$$

22. Given that $(1 + x + x^2)^n = a_0 + a_1 x + a_2 x^2 + \dots + a_{2n} x^{2n}$, find the values of :

(i) $a_0 + a_1 + a_2 + \dots + a_{2n}$; (ii) $a_0 - a_1 + a_2 - a_3 + \dots + a_{2n}$; (iii) $a_0^2 - a_1^2 + a_2^2 - a_3^2 + \dots + a_{2n}^2$

दिया गया है $(1 + x + x^2)^n = a_0 + a_1 x + a_2 x^2 + \dots + a_{2n} x^{2n}$, तो निम्न के मान ज्ञात कीजिए –

(i) $a_0 + a_1 + a_2 + \dots + a_{2n}$; (ii) $a_0 - a_1 + a_2 - a_3 + \dots + a_{2n}$; (iii) $a_0^2 - a_1^2 + a_2^2 - a_3^2 + \dots + a_{2n}^2$

Ans. (i) 3^n (ii) 1, (iii) a_n

Sol. $(1 + x + x^2)^n = a_0 + a_1 x + a_2 x^2 + \dots + a_{2n} x^{2n}$

$$(i) a_0 + a_1 + a_2 + \dots + a_{2n} = (1 + 1 + 1)^n = 3^n$$

$$(ii) a_0 - a_1 + a_2 - a_3 + \dots + a_{2n} = (1 - 1 + (-1)^2)^n = 1$$

$$(iii) a_0^2 - a_1^2 + a_2^2 - a_3^2 + \dots + a_{2n}^2 =$$

$$\therefore (1 + x + x^2)^n = a_0 + a_1 x + a_2 x^2 + \dots + a_{2n} x^{2n} \quad \text{_____ (i)}$$

$$x \rightarrow \frac{1}{x}$$

$$\left(1 + \frac{1}{x} + \frac{1}{x^2}\right)^n = a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + \dots + \frac{a_{2n}}{x^{2n}}$$

$$\left[\frac{(1 + x + x^2)^n}{x^{2n}}\right] = \frac{a_0 x^{2n} + a_1 x^{2n-1} + a_2 x^{2n-2} + \dots + a_{2n}}{x^{2n}}$$

$$x \rightarrow -x$$

$$\left[\frac{(1 - x + x^2)^n}{x^{2n}}\right] = \frac{a_0 x^{2n} - a_1 x^{2n-1} + a_2 x^{2n-2} + \dots + a_{2n}}{x^{2n}} \quad \text{_____ (ii)}$$

By (i) $\times$ (ii); we get

$$\Rightarrow \left[(1 + x^2)^2 - x^2\right]^n = \dots + x^{2n} (a_0^2 - a_1^2 + a_2^2 - a_3^2 + \dots + a_{2n}^2)$$

Comparing coefficient of x^{2n} in $[1 + x^4 + x^2]^n$

$$\therefore (1 + x + x^2)^n = a_0 + a_1 x + \dots + a_{2n} x^{2n}$$

$$x \rightarrow x^2$$

$$(1 + x^2 + x^4)^n = a_0 + a_1 x^2 + \dots + a_n x^{2n} + \dots + a_{2n} x^{4n}$$

$$\Rightarrow a_0^2 - a_1^2 + a_2^2 - a_3^2 + \dots + a_{2n}^2 = a_n$$

23. Prove that $\sum_{K=0}^n {}^n C_K \sin Kx \cdot \cos(n-K)x = 2^{n-1} \sin nx$.

सिद्ध कीजिए $\sum_{K=0}^n {}^n C_K \sin Kx \cdot \cos(n-K)x = 2^{n-1} \sin nx$

Ans.

Sol. $S = {}^nC_0 \sin(0x) \cos nx + {}^nC_1 \sin x \cos(n-1)x$

$$+ \dots + {}^nC_n \sin nx \cos(0x) \dots \quad (1)$$

$$S = {}^nC_0 \sin nx \cos(0x) + {}^nC_1 \sin(n-1)x \cos x +$$

$$+ \dots + {}^nC_n \sin(0x) \cos nx \dots \quad (2)$$

Add (1) & (2)

$$2S = ({}^nC_0 + {}^nC_1 + \dots + {}^nC_n) \sin nx \Rightarrow S = 2^{n-1} \sin nx$$

24. Find the number of divisors of the number

$$N = {}^{2000}C_1 + 2 \cdot {}^{2000}C_2 + 3 \cdot {}^{2000}C_3 + \dots + 2000 \cdot {}^{2000}C_{2000}$$

$$N = {}^{2000}C_1 + 2 \cdot {}^{2000}C_2 + 3 \cdot {}^{2000}C_3 + \dots + 2000 \cdot {}^{2000}C_{2000} \text{ के विभाजकों की संख्या ज्ञात कीजिए।}$$

Ans. 8016

Sol. $N = \sum_{r=0}^{2000} {}^{2000}C_r \cdot r$

$$= \sum_{r=1}^{2000} \frac{{}^{2000}C_r}{r} \cdot r \cdot 1999 \cdot {}^{1999}C_{r-1}$$

$$= 2000 \sum_{r=1}^{2000} {}^{1999}C_{r-1}$$

$$= 2 \times 10^3 \times (2)^{1999} = 2^{2000} \cdot 2^3 \cdot 5^3 = 2^{2003} \cdot 5^3$$

$$\Rightarrow \text{No. of divisor} = 2004 \times 4 = 8016$$

25. Prove the following identities using the theory of permutation where $C_0, C_1, C_2, \dots, C_n$ are the combinatorial coefficients in the expansion of $(1+x)^n$, $n \in \mathbb{N}$, then prove the following:

क्रमचय सिद्धान्त का प्रयोग कर सर्वसमिकाओं को सिद्ध कीजिये। यदि $C_0, C_1, C_2, \dots, C_n, (1+x)^n, n \in \mathbb{N}$ के प्रसार में द्विपद गुणांक है। तब निम्न को सिद्ध कीजिए –

$$(a) C_0^2 + C_1^2 + C_2^2 + \dots + C_n^2 = \frac{(2n)!}{n! n!}$$

$$(b) C_0 C_1 + C_1 C_2 + C_2 C_3 + \dots + C_{n-1} C_n = \frac{(2n)!}{(n+1)! (n-1)!}$$

$$(c) C_0 C_r + C_1 C_{r+1} + C_2 C_{r+2} + \dots + C_{n-r} C_n = \frac{2n!}{(n-r)! (n+r)!}$$

$$(d) \sum_{r=0}^{n-2} ({}^nC_r \cdot {}^nC_{r+2}) = \frac{(2n)!}{(n-2)! (n+2)!}$$

Ans.

Sol. (a) $C_0^2 + C_1^2 + C_2^2 + \dots + C_n^2$

$$(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$$

$$(x+1)^n = C_0x^n + C_1x^{n-1} + \dots + C_n$$

$$\Rightarrow (1+x)^{2n} = \left[\sum_{r=0}^n C_r x^r \right] \left[\sum_{r=0}^n C_r x^{n-r} \right]$$

Comparing coefficient of x^n in $(1+x)^{2n}$

$$\Rightarrow C_0^2 + C_1^2 + C_2^2 + \dots + C_n^2 = {}^{2n}C_n = \frac{|2n|}{|n|n}$$

(b) $C_0C_1 + C_1C_2 + C_2C_3 + \dots + C_{n-1}C_n$

Comparing coefficient of x^{n-1} in $(1+x)^{2n}$

$$\Rightarrow C_0C_1 + C_1C_2 + \dots + C_{n-1}C_n = {}^{2n}C_{n-1} = \frac{|2n|}{|n-1|n+1}$$

(c) We know that

$$(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_2x^r + \dots + C_nx^n \quad \text{_____ (i)}$$

$$\left(1 + \frac{1}{x}\right)^n = C_0 + C_1 \frac{1}{x} + C_2 \frac{1}{x^2} + \dots + C_r \frac{1}{x^r} + \dots + C_n \frac{1}{x^n} \quad \text{_____ (ii)}$$

Multiply (i) & (ii) we get

$$\frac{(1+x)^{2n}}{x^n} = (C_0 + C_1x + \dots + C_r x^r + \dots + C_n x^n) \left(C_0 + \frac{C_1}{x} + \frac{C_2}{x^2} + \frac{C_n}{x^n} \right) \quad \text{_____ (iii)}$$

Now coefficient of $\frac{1}{x^r}$ in RHS

$$= C_0C_r + C_1C_{r+1} + \dots + C_{n-r}C_n$$

$\therefore$ Coefficient of $\frac{1}{x^r}$ in LHS

$$= \text{coefficient of } x^{n-r} \text{ in } (1+x)^{2n} = {}^{2n}C_{n-r}$$

$$= \frac{2n!}{(n-r)!(n+r)!}$$

But (iii) is an identity

$\therefore$ Coefficient of $\frac{1}{x^r}$ in RHS

= coefficient of $\frac{1}{x^r}$ in LHS

$$\Rightarrow C_0 C_r + C_1 C_{r+1} + \dots + C_{n-r} C_n = \frac{2n!}{(n-r)!(n+r)!}$$

$$(d) \sum_{r=0}^{n-2} {}^n C_r {}^n C_{r+2} = {}^2 C_0 {}^n C_2 + {}^n C_1 {}^n C_3 + {}^n C_2 {}^n C_4 + \dots + {}^n C_{n-2} {}^n C_n$$

Comparing coefficient of x^{n-2} in

$$\sum_{r=0}^{n-2} {}^n C_r {}^n C_{r+2} = \frac{|2n|}{|n-2|n+2}$$

26. Prove the following identities using the theory of permutation where $C_0, C_1, C_2, \dots, C_n$ are the combinatorial coefficients in the expansion of $(1+x)^n$, $n \in N$, then prove the following:

क्रमचय सिद्धान्त का प्रयोग कर सर्वसमिकाओं को सिद्ध कीजिये। यदि $C_0, C_1, C_2, \dots, C_n, (1+x)^n, n \in N$ के प्रसार में द्विपद गुणांक है। तब निम्न को सिद्ध कीजिए।

$$(a) C_1 + 2C_2 + 3C_3 + \dots + n \cdot C_n = n \cdot 2^{n-1}$$

$$(b) C_0 + 2C_1 + 3C_2 + \dots + (n+1)C_n = (n+2)2^{n-1}$$

$$(c) C_0 + 3C_1 + 5C_2 + \dots + (2n+1)C_n = (n+1)2^n$$

$$(d) (C_0 + C_1)(C_1 + C_2)(C_2 + C_3) \dots (C_{n-1} + C_n) = \frac{C_0 \cdot C_1 \cdot C_2 \dots C_{n-1} (n+1)^n}{n!}$$

$$(e) 1 \cdot C_0^2 + 3 \cdot C_1^2 + 5 \cdot C_2^2 + \dots + (2n+1) C_n^2 = \frac{(n+1)(2n)!}{n! n!}$$

Ans.

Sol. $(1+x)^n = {}^n C_0 + {}^n C_1 x + \dots + {}^n C_n x^n$

$$(a) \sum_{r=1}^n r \cdot {}^n C_r = \sum_{r=1}^n r \cdot \frac{n \cdot {}^{n-1} C_{r-1}}{r}$$

$$= n \left[{}^{n-1} C_0 + {}^{n-1} C_1 + \dots + {}^{n-1} C_{n-1} \right]$$

$$= n \cdot 2^{n-1}$$

$$(b) C_0 + 2C_1 + 3C_2 + \dots + (n+1)C_n$$

$$= \sum_{r=0}^n (r+1)^n C_r = \sum_{r=0}^n r \cdot {}^n C_r + \sum_{r=0}^n {}^n C_r$$

$$= n \cdot 2^{n-1} + 2^n = 2^{n-1} (n+2)$$

$$(c) C_0 + 3C_1 + 5C_2 + \dots + (2n+1) C_n$$

$$= \sum_{r=0}^n (2r+1) {}^n C_r$$

$$= 2 \sum_{r=0}^n r \cdot {}^n C_r + \sum_{r=0}^n {}^n C_r$$

$$= 2^n \cdot 2^{n-1} + 2^n$$

$$= 2^n (n+1)$$

$$(d) (C_0 + C_1)(C_1 + C_2) \dots (C_{n-1} + C_n)$$

$$= ({}^{n+1}C_1)({}^{n+1}C_2) \dots ({}^{n+1}C_n) = ((n+1) \cdot C_0) \left(\frac{n+1}{2} \cdot C_1 \right) \dots \left(\frac{n+1}{n} \cdot C_{n-1} \right)$$

$$= \frac{(n+1)^n}{n!} C_0 C_1 \dots C_{n-1}$$

$$(e) 1 \cdot C_0^2 + 3 \cdot C_1^2 + 5 \cdot C_2^2 + \dots + (2n+1) C_n^2$$

$$= \sum_{r=0}^n (2r+1) C_r^2$$

$$= 2 \sum_{r=0}^n r \cdot C_r^2 + \sum_{r=0}^n C_r^2$$

$$= 2 \sum_{r=0}^n r \cdot \frac{n}{r} {}^{n-1} C_{r-1} \cdot {}^n C_r + \sum_{r=0}^n ({}^n C_r)^2$$

$$= 2n \cdot \sum_{r=1}^n {}^{n-1} C_{r-1} \cdot {}^n C_r + \sum_{r=0}^n ({}^n C_r)^2$$

$$= 2n \left[\underbrace{{}^{n-1} C_0}_{\text{Coefficient of } x^{n-1}} \underbrace{{}^n C_1}_{\text{Coefficient of } x} + {}^{n-1} C_1 {}^n C_2 + \dots + {}^{n-1} C_{n-1} {}^n C_n \right] + \sum_{r=0}^n ({}^n C_r)^2$$

$$= 2n \cdot \text{coefficient of } x^n \text{ in } (1+x)^{2n-1} + \frac{|2n|}{|n|n}$$

$$\begin{aligned}
&= 2n \cdot \frac{|2n-1|}{|n-1|n} + \frac{|2n|}{|n|n} \\
&= 2n \left[\frac{|2n-1|}{|n-1|n} + \frac{|2n-1|}{|n|n} \right] = \frac{|2n|}{|n|n-1} \left[1 + \frac{1}{n} \right] \\
&= \frac{|2n|}{|n|n} (n+1)
\end{aligned}$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 27 to 29

A path of length n is a sequence of points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ with integer coordinates such that for all i between 1 and $n-1$ both inclusive,

either $x_{i+1} = x_i + 1$ and $y_{i+1} = y_i$ (in which case we say the i^{th} step is rightward)

or $x_{i+1} = x_i$ and $y_{i+1} = y_i + 1$ (in which case we say that the i^{th} step is upward).

This path is said to start at (x_1, y_1) and end at (x_n, y_n) . Let $P(a, b)$, for a and b non-negative integers, denotes the number of paths that start at $(0, 0)$ and end at (a, b) .

n लम्बाई का पाथ एक अनुक्रम है जिसके बिन्दु पूर्णांक निर्देशांकों $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ है। इस प्रकार की 1 और $n-1$ के मध्य, जहाँ 1 और $n-1$ भी शामिल है, सभी i के लिए

या तो $x_{i+1} = x_i + 1$ और $y_{i+1} = y_i$ (में i^{th} स्टेप दांयी ओर है)

या $x_{i+1} = x_i$ और $y_{i+1} = y_i + 1$ (में i^{th} स्टेप ऊपर की ओर है)

इस पथ का प्रारम्भिक बिन्दु (x_1, y_1) और अन्त बिन्दु (x_n, y_n) है। माना $P(a, b)$, जहाँ a, b अऋणात्मक पूर्णांक है। उन पाथों की संख्या दर्शाते हैं जिसका प्रारम्भिक बिन्दु $(0, 0)$ और अन्त बिन्दु (a, b) है।

27. The value of $\sum_{i=0}^{10} P(i, 10-i)$, is :

$\sum_{i=0}^{10} P(i, 10-i)$ का मान होगा:

- (A) 1024 (B) 512 (C) 256 (D) 128

Ans. A

Sol. $\sum_{i=0}^{10} P(i, 10-i) = \sum_{i=0}^{10} \frac{|10|}{|i|10-i}$

$$= \sum_{i=0}^{10} {}^{10}C_i = 2^{10} = 1024$$

28. Number of ordered pairs (i, j) where $i \neq j$ for which $P(i, 100-i) = P(j, 100-j)$, is :

$P(i, 100-i) = P(j, 100-j)$ के लिए क्रमित युग्मों (i, j) की संख्या (जहाँ $i \neq j$) है:

(A) 50

(B) 99

(C) 100

(D) 101

Ans. C**Sol.** $i \neq j \quad P(i, 100 - i) = P(j, 100 - j)$

$$\neq {}^{100}C_i = {}^{100}C_j$$

$$\Rightarrow i = j \text{ (wrong)} \quad \text{or} \quad \underbrace{i + j = 100}_{\substack{(0,100) \\ (1,99) \\ (50,50) \\ (49,51) \\ (99,1) \\ (100,0)}} \quad \& i \neq j$$

$$\Rightarrow \text{No. of ordered pair} = 101 - 1 = 100$$

29. The sum $P(43, 4) + \sum_{j=1}^5 P(49 - j, 3)$ is equal to :

$$P(43, 4) + \sum_{j=1}^5 P(49 - j, 3) \text{ का योग बराबर :}$$

(A) $P(4, 48)$ (B) $P(3, 49)$ (C) $P(4, 47)$ (D) $P(5, 47)$ **Ans.** A

$$\text{Sol. } P(43, 4) + \sum_{j=1}^5 P(49 - j, 3)$$

$$= \frac{|47|}{|43|4} + \sum_{j=1}^5 \frac{|52-j|}{|49-j|3}$$

$$= {}^{47}C_4 + \sum_{j=1}^5 {}^{52-j}C_3$$

$$= {}^{47}C_4 + {}^{47}C_3 + {}^{48}C_3 + {}^{49}C_3 + {}^{50}C_3 + {}^{51}C_3$$

$$= {}^{48}C_4 + {}^{48}C_3 + {}^{49}C_3 + {}^{50}C_3 + {}^{51}C_3$$

$$= {}^{52}C_4$$

$$= \frac{|52|}{|4|48}$$

$$= P(4, 48)$$

Paragraph for Question Nos. 30 to 32

$$\text{Let } S_k = 3^k \binom{100}{0} \binom{100}{k} - 3^{k-1} \binom{100}{1} \binom{99}{k-1} + 3^{k-2} \binom{100}{2} \binom{98}{k-2} - \dots + (-1)^k \binom{100}{k} \binom{100-k}{0}$$

where $\binom{n}{r}$ represents the coefficient of x^r in the binomial expansion of $(1+x)^n \forall n \in \mathbb{N}$

$$\text{and } V_k = \left(\frac{1}{2}\right)^k S_k = M(100, k).$$

$$\text{माना } S_k = 3^k \binom{100}{0} \binom{100}{k} - 3^{k-1} \binom{100}{1} \binom{99}{k-1} + 3^{k-2} \binom{100}{2} \binom{98}{k-2} - \dots + (-1)^k \binom{100}{k} \binom{100-k}{0}$$

जहाँ $\binom{n}{r} (1+x)^n \forall n \in \mathbb{N}$ के प्रसार में x^r का गुणांक है और $V_k = \left(\frac{1}{2}\right)^k S_k = M(100, k).$

30. The value of $\sum_{k=0}^{100} S_k S_{100-k}$ equals :

$$\sum_{k=0}^{100} S_k S_{100-k} \text{ का मान है :}$$

$$(A) \binom{200}{100} \quad (B) 2^{100} \binom{200}{100} \quad (C) 2^{200} \binom{200}{100} \quad (D) 2^{99} \binom{200}{99}$$

Ans. B

$$\text{Sol. } S_k = 3^k {}^{100}C_0 {}^{100}C_k - 3^{k-1} {}^{100}C_1 {}^{99}C_{k-1} + \dots + (-1)^k \cdot {}^{100}C_k {}^{100-k}C_0$$

$$(1+x)^{100} = {}^{100}C_0 + {}^{100}C_1 x + \dots + {}^{100}C_k x^k + \dots + {}^{100}C_{100} x^{100}$$

$$S_K = \sum_{r=0}^K 3^{K-r} {}^{100}C_r {}^{100-r}C_{K-r} (-1)^r$$

$$= \sum_{r=0}^K 3^{K-r} \frac{{}^{100}C_r}{{}^{100-r}C_r} \frac{{}^{100-r}C_{K-r}}{{}^{100-K}C_{K-r}} (-1)^r$$

$$= \sum_{r=0}^K 3^{K-r} (-1)^r {}^{100}C_K {}^K C_r$$

$$= {}^{100}C_K (3-1)^K = 2^K \cdot {}^{100}C_K$$

$$\Rightarrow V_K = \frac{1}{2^K} S_K = {}^{100}C_K = M(100, K)$$

$$\begin{aligned}
\sum_{K=0}^{100} S_K S_{100-K} &= \sum_{K=0}^{100} {}^{100}C_K \cdot {}^{100}C_{100-K} \cdot 2^K \cdot 2^{100-K} \\
&= 2^{100} \sum_{K=0}^{100} \left({}^{100}C_K \right)^2 \rightarrow \text{Coefficient of } x^{100} \text{ in } (1+x)^{200} \\
&= 2^{100} \cdot {}^{200}C_{100} \\
&= 2^{100} \cdot \binom{200}{100}
\end{aligned}$$

31. The value of the $M(100, 49) + M(100, 50)$ is equal to :

$M(100, 49) + M(100, 50)$ का मान है :

- (A) $M(99, 50)$ (B) $M(100, 51)$ (C) $M(101, 49)$ (D) $M(101, 51)$

Ans. D

Sol. $\because M(100, K) = {}^{100}C_K$

$$\Rightarrow M(100, 49) + M(100, 50)$$

$$= {}^{100}C_{49} + {}^{100}C_{50}$$

$$= {}^{100}C_{50} = M(101, 50) = M(101, 51)$$

32. In an examination, there are 100 true-false type questions. The number of ways in which one student can make his choice to attempt or not to attempt the questions is given by :

किसी परीक्षा में 100 सत्य असत्य प्रकार के प्रश्न हैं। तब कोई विद्यार्थी कुल कितने प्रकार से प्रश्न का उत्तर दे या नहीं दे सकता है :

- (A) $\sum_{k=0}^{100} S_k$ (B) $\sum_{k=0}^{100} V_k$ (C) $\sum_{k=1}^{100} S_k$ (D) $\sum_{k=1}^{100} V_k$

Ans. A

Sol. $100 \rightarrow \text{question (True/False)}$

$$= {}^{100}C_0 \cdot 2^0 + {}^{100}C_1 \cdot 2 + \dots + {}^{100}C_{100} \cdot 2^{100}$$

$$= \sum_{K=0}^{100} {}^{100}C_K \cdot 2^K = \sum_{K=0}^{100} S_K$$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

33. In the expansion of $\left(x - \frac{\alpha}{x}\right)^n$ and $\left(x + \frac{\beta}{x^2}\right)^n$ in powers of x have one term independent of x , then n is divisible by :

यदि $\left(x - \frac{\alpha}{x}\right)^n$ तथा $\left(x + \frac{\beta}{x^2}\right)^n$ के प्रसार में x की घातों के रूप में केवल एक पद ही x से स्वतन्त्र हो तब n विभाजित होगा:

- (A) 2 (B) 3 (C) 4 (D) 6

Ans. ABD

Sol. $\left(x - \frac{\alpha}{x}\right)^n = \sum_{r=0}^n {}^nC_r x^{n-2r} \cdot (-\alpha)^r \rightarrow$ Term independent of x if $n = 2r$

$$\left(x + \frac{\beta}{x^2}\right)^n = \sum_{r=0}^n {}^nC_r x^{n-3r} \cdot (\beta)^r \rightarrow \text{Term independent of } x \text{ if } n = 3r$$

$\Rightarrow n$ is divisible by 2, 3 or 6.

34. In the expansion of $(a + b + c)^{10}$:

- (A) total number of terms is 66 (B) coefficient of $a^8 bc$ is 90
(C) coefficient of $a^4 b^5 c^3$ is 0 (D) none of these

$\therefore$ इ. बद्ध के प्रसार में :

- (A) कुल पदों की संख्या 66 है। (B) $a^8 bc$ का गुणांक 90 है।
(C) $a^4 b^5 c^3$ का गुणांक 0 है। (D) इनमें से कोई नहीं

Ans. ABC

Sol. $(a + b + c)^{10}$

$$\text{Total terms are } {}^{10+3-1}C_2 = {}^{12}C_2 = \frac{12 \times 11}{2} = 66$$

$$(a + b + c)^{10} = \sum_{P_1+P_2+P_3=10} \frac{10!}{P_1!P_2!P_3!} a^{P_1} b^{P_2} c^{P_3}$$

$$\text{Coefficient of } a^8 bc \text{ is } \frac{10!}{8!1!1!} = 90$$

$$\text{Coefficient of } a^4 b^5 c^3 \text{ is } 0 \text{ as } P_1 + P_2 + P_3 = 12 \neq 10$$

35. In the expansion of $\left(x + \frac{\alpha}{x^2}\right)^n$, $\alpha \neq 0$, if no term is independent of x , then n is :

$\left(x + \frac{a}{x^2}\right)^n$, $a \neq 0$ के प्रसार में कोई भी पद x से स्वतन्त्र नहीं है तब $n =$

- (A) 10 (B) 12 (C) 16 (D) 20

Ans. ACD

Sol. $\left(x + \frac{a}{x^2}\right)^n$; $a \neq 0 = \sum_{r=0}^n x^{n-3r} \cdot {}^nC_r \cdot a^r$

If no term is independent of x then

$$n - 3r \neq 0$$

$$\Rightarrow n \neq 3r$$

$$\Rightarrow n \neq 0, 3, 6, 9, 12, 15, \dots$$

$$\Rightarrow n = 10, 16 \text{ \& } 20$$

36. In the expansion of $\left(x + \frac{1}{x} + 2\right)^{2009}$

- (A) Middle term is ${}^{4018}C_{2009}$
 (B) The term independent of x is ${}^{4018}C_{2009}$
 (C) Coefficient of x in the expansion is ${}^{4018}C_{2010}$
 (D) Coefficient of $1/x$ in the expansion is ${}^{4018}C_{2008}$

$\left(x + \frac{1}{x} + 2\right)^{2009}$ के प्रसार में –

- (A) मध्य पद ${}^{4018}C_{2009}$ है। (B) x से स्वतंत्र पद ${}^{4018}C_{2009}$ है।
 (C) x का गुणांक ${}^{4018}C_{2010}$ है। (D) $1/x$ का गुणांक ${}^{4018}C_{2008}$ है।

Ans. ABCD

Sol. $\left(x + \frac{1}{x} + 2\right)^{2009}$

$$= \left(\frac{x^2 + 2x + 1}{x}\right)^{2009} = \frac{1}{x^{2009}} (x+1)^{4018}$$

$$= 1 \left[\sum_{r=0}^{4018} {}^{4018}C_r x^{r-2009} \right]$$

Term independent of x if $r = 2009$

$\Rightarrow$ Coefficient of this is ${}^{4018}C_{2009}$.

Coefficient of $x = {}^{4018}C_{2010}$

Coefficient of $\frac{1}{x} = {}^{4018}C_{2008}$

Coefficient of middle term is ${}^{4018}C_{2009}$

37. Which of the following statement(s) is/are correct?

(A) $1 + \frac{2}{2} + \frac{3}{2^2} + \frac{4}{2^3} + \dots + \infty = 4$

(B) Integral part of $(9 + 4\sqrt{5})^n$, is even, $n \in \mathbb{N}$.

(C) $({}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n)^2 = 1 + {}^{2n}C_1 + {}^{2n}C_2 + \dots + {}^{2n}C_{2n}$

(D) $\frac{1}{(3+2x)^2}$ can be expanded as infinite series in ascending powers of x only if $|x| < \frac{2}{3}$

निम्न में से कौनसा/से कथन सत्य है ?

(A) $1 + \frac{2}{2} + \frac{3}{2^2} + \frac{4}{2^3} + \dots + \infty = 4$

(B) $(9 + 4\sqrt{5})^n$ का पूर्णांक मान सम है, जहाँ $n \in \mathbb{N}$.

(C) $({}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n)^2 = 1 + {}^{2n}C_1 + {}^{2n}C_2 + \dots + {}^{2n}C_{2n}$

(D) यदि $|x| < \frac{2}{3}$ तो $\frac{1}{(3+2x)^2}$ को x के बढ़ते घात में अनन्त पदों तक लिखा जा सकता है

Ans. AC

Sol. A) $\left(1 - \frac{1}{2}\right)^{-2} = \left(\frac{1}{2}\right)^{-2} = \frac{1}{2^{-2}} = 4 \Rightarrow$ (A) is correct
 (B) it should be odd $\Rightarrow$ (B) is incorrect
 (C) L.H.S. $= (2^n)^2$; R.H.S. $= 2^{2n} \Rightarrow$ (C) is correct
 (D) $|x| < \frac{3}{2} \Rightarrow$ (D) is incorrect

38. If $(9 + \sqrt{80})^n = 1 + f$ where I, n are integers and $0 < f < 1$, then
 (A) I is an odd integer (B) I is an even integer
 (C) $(I + f)(1 - f) = 1$ (D) $1 - f = (9 - \sqrt{80})^n$

यदि $(9 + \sqrt{80})^n = I + f$ जहाँ I तथा n पूर्णांक है जहाँ $0 < f < 1$ तो

(A) I विषम पूर्णांक

(B) I सम पूर्णांक है

(C) $(I + f)(1 - f) = 1$

(D) $1 - f = (9 - \sqrt{80})^n$

Ans. ACD

Sol. $(9 + 4\sqrt{5})^n = 3 + f$

let $(9 - 4\sqrt{5})^n = f'$

$I + f + f' = (9 + 4\sqrt{5})^n + (9 - 4\sqrt{5})^n = \text{even integer}$

$0 < f + f' < 2 \Rightarrow f + f' = 1$

$\therefore I$ is an odd integer $\Rightarrow$ (A)

39. Let $(1 + x^2)^2 (1 + x)^n = A_0 + A_1 x + A_2 x^2 + \dots$. If A_0, A_1, A_2 are in A.P. then the value of n is :

माना $(1 + x^2)^2 (1 + x)^n = A_0 + A_1 x + A_2 x^2 + \dots$ यदि A_0, A_1, A_2 समान्तर श्रेणी में हो, तो n बराबर होगा :

(A) 2

(B) 3

(C) 5

(D) 7

Ans. AB

Sol. $(1 + x^2)^2 (1 + x)^n = (1 + 2x^2 + x^4) ({}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots)$
 $= {}^nC_0 + {}^nC_1 x + ({}^nC_2 + 2 \cdot {}^nC_0) x^2 + \dots$

Hence $A_0 = 1$; $A_1 = {}^nC_1$; $A_2 = {}^nC_2 + 2$ which are in A.P.

40. If ${}^{100}C_6 + 4 \cdot {}^{100}C_7 + 6 \cdot {}^{100}C_8 + 4 \cdot {}^{100}C_9 + {}^{100}C_{10}$ has the value equal to xC_y ; then the possible value(s) of $x + y$ can be :

यदि ${}^{100}C_6 + 4 \cdot {}^{100}C_7 + 6 \cdot {}^{100}C_8 + 4 \cdot {}^{100}C_9 + {}^{100}C_{10}$ का मान xC_y के बराबर है ; तो $x + y$ के संभव मान हो सकते हैं :

(A) 112

(B) 114

(C) 196

(D) 198

Ans. BD

Sol. ${}^{100}C_6 + 4 \cdot {}^{100}C_7 + 6 \cdot {}^{100}C_8 + 4 \cdot {}^{100}C_9 + {}^{100}C_{10}$
 $= ({}^{100}C_7 + {}^{100}C_6) + 3 ({}^{100}C_7 + {}^{100}C_8) + 3 ({}^{100}C_8 + {}^{100}C_9) + {}^{100}C_9 + {}^{100}C_{10}$
 $= {}^{101}C_7 + 3 \cdot {}^{101}C_8 + 3 \cdot {}^{101}C_9 + {}^{101}C_{10}$
 $= ({}^{101}C_7 + {}^{101}C_8) + 2 [{}^{101}C_8 + {}^{101}C_9] + [{}^{101}C_{10} + {}^{101}C_9]$
 $= {}^{102}C_8 + 2 \cdot {}^{102}C_9 + {}^{102}C_{10}$
 $= {}^{103}C_9 + {}^{103}C_{10}$
 $= {}^{104}C_{10} = {}^{104}C_{94}$
 $\Rightarrow x + y = 114$ or $x + y = 198$

MATCH THE COLUMN TYPE QUESTION(S)

41. Match the column

Column I

Column II

(A) If the coefficients of x^7 and x^8 in $\left(2 + \frac{x}{3}\right)^n$ are equal, then n is

(P) 2

(B) If the coefficient of x in the expansion of $\left(x - \frac{1}{ax^2}\right)^{10}$ is -15 ,

then the value of a , is

(Q) 01

(C) The remainder when 2^{2003} is divided by 17 is

(R) 55

(D) The last two digits of 3^{400} , are

(S) 8

Column I

Column II

(A) यदि $\left(2 + \frac{x}{3}\right)^n$ के प्रसार में x^7 व x^8 के गुणांक बराबर हैं तब $n =$

(P) 2

(B) $\left(x - \frac{1}{ax^2}\right)^{10}$ के प्रसार में x का गुणांक -15 है तब $a =$

(Q) 01

(C) 2^{2003} को 17 से विभाजित करने पर शेषफल

(R) 55

; व 3^{400} के अन्तिम दो अंक

(S) 8

Ans. (A) R (B) P (C) S (D) Q

Sol. (A) Coefficient of x^7 & x^8 are equal in $\left(2 + \frac{x}{3}\right)^n$

$$\Rightarrow {}^nC_7 2^{n-7} \cdot \frac{x^7}{3^7} = {}^nC_8 2^{n-8} \cdot \frac{x^8}{3^8}$$

$$\Rightarrow \frac{{}^nC_7 2^n \cdot \frac{1}{2^7} \cdot \frac{1}{3^7}}{{}^nC_8 2^n \cdot \frac{1}{2^8} \cdot \frac{1}{3^8}} = \frac{{}^nC_7}{{}^nC_8} \cdot \frac{2}{3}$$

$$\Rightarrow \frac{1}{(n-7)} = \frac{1}{68}$$

$$\Rightarrow n - 7 = 48$$

$$\Rightarrow n = 55$$

$$(B) \left(x - \frac{1}{ax^2}\right)^{10} = \sum_{r=0}^{10} {}^{10}C_r x^{10-r} \cdot \frac{(-1)^r}{a^r \cdot x^{2r}}$$

$$= \sum_{r=0}^{10} {}^{10}C_r \cdot x^{10-3r} \cdot \left(\frac{-1}{a}\right)^r$$

$$\text{For } x^1; \quad 10 - 3r = 1$$

$$3r = 9$$

$$r = 3$$

$$\Rightarrow {}^{10}C_3 \left(\frac{-1}{a} \right)^3 = -15$$

$$\Rightarrow \frac{10 \times 9 \times 8}{6} = 15a^3$$

$$\Rightarrow 9 \times 8 \times 2 = 15a^3$$

$$\Rightarrow a^3 = 8$$

$$\Rightarrow a = 2$$

$$(C) 2^{2003} = 2 \cdot 2^{2002} = 2^3 \cdot (2)^{2000} = 8 \cdot (16)^{500}$$

$$= 8 \cdot [17 - 1]^{500}$$

$$= 8 \sum_{r=0}^{500} {}^{500}C_r 17^{500-r} (-1)^r$$

$$= 17\lambda + {}^{8500}C_{500}$$

$$\therefore \text{Remainder} = 8$$

$$(D) 3^{400} = (81)^{100} = (80 + 1)^{100}$$

$$= \sum_{r=0}^{100} {}^{100}C_r (80)^r$$

$$= {}^{100}C_0 + {}^{100}C_1 \cdot 80 + 100\lambda$$

$$= {}^{100}C_0 + 100\lambda'; \lambda \in \mathbb{Z}$$

$$= 100\lambda' + 1$$

$$\therefore \text{Last digit will be 01.}$$